Diversified Trend Following: Many Bets on the Sharpe Ratio, Few on Convexity*

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Abstract

Investors hold trend-following strategies for their convexity: positively skewed returns, often high when equities fall. This article shows what sets this convexity when a trend follower trades many futures markets, by splitting its portfolio into eigen-trends, trend followers on principal components. In a Gaussian benchmark, the correlation spectrum sets how much single-market skewness survives, and the directions shift the crisis payoff, not skewness. On 81 futures markets, 2001–2026, diversification creates 18 bets on the Sharpe ratio but only 4 on convexity: one risk-on/risk-off trend holds half the variance and three quarters of the convexity implied by the spectrum and, with the next two, earns most of the crisis payoff. In stress, the 4 bets narrow to 3. Filters built on estimated trend premia, including supervised (SPCA) and risk-premium (RP-PCA) principal component analysis, span a trade-off among skewness, Sharpe ratio, and crisis payoff; net of costs, they differ in which of these they keep, not in a significant Sharpe gain. The risk rule decides how much convexity is measured: daily position-risk targeting cuts 55-day skewness from 1.38 to 0.07. A trend follower can diversify its Sharpe ratio across markets, but its convexity remains a bet on a few common trends.

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1. Introduction

Investors hold trend-following strategies, such as those of commodity trading advisors (CTAs) and managed-futures funds, for their convexity: their returns are positively skewed and tend to be high when equity markets fall (Fung and Hsieh, 2001; Darius, Ilhan, Mulvey, Simsek, and Sircar, 2002; Potters and Bouchaud, 2006; Moskowitz, Ooi, and Pedersen, 2012; Hurst, Ooi, and Pedersen, 2017). This makes trend following an exception to the rule that risk premia with higher Sharpe ratios are more negatively skewed (Lempérière, Deremble, Nguyen, Seager, Potters, and Bouchaud, 2017). In this article, convexity means positive skewness of trend-following profit and loss (P&L) over a horizon; crisis payoff and drawdowns are reported separately. For a single market, the source of convexity is well understood: positions grow after gains, so risk rises after gains and the P&L becomes positively skewed (Martin and Zou, 2012; Martin, 2023; Sepp and Lucic, 2026).

Trend followers, however, trade many markets, and these markets share trends in risk appetite, interest rates, and commodity demand. The usual measures of diversification count markets or effective bets on variance (Grinold, 1989; Meucci, 2009; Roncalli and Weisang, 2016), so they describe the Sharpe ratio, not convexity. Diversification lowers skewness (Simkowitz and Beedles, 1978), which is harder to diversify than variance (Kim, Kim, Kim, and Bang, 2018), and the convexity of trend P&L is diluted across markets (Dao, Nguyen, Deremble, Lempérière, Bouchaud, and Potters, 2017; Capital Fund Management, 2018; Firoozye and Koshiyama, 2020). The data pose a puzzle. A single futures trend follower has a median 55-day skewness of 2.2. The same trend follower on all 81 markets, at equal risk, still has 1.25, far more than the 0.24 that independent markets would leave and the 0.51 that its 18 independent bets on the Sharpe ratio would imply, while its Sharpe ratio rises from 0.23 to 0.99. This article studies three questions: how many independent bets a diversified trend follower holds on convexity and where they sit, whether its allocation across markets can improve the trade-off between the Sharpe ratio and convexity, and how much

convexity its risk rule preserves.

The central finding is that the convexity of a diversified trend follower sits in a few common trends, and its crisis payoff in the same few, led by a risk-on/risk-off trend on a portfolio long equities and short government bonds. A common trend here is a portfolio through which many markets move together, a principal component of their correlation matrix, not a stochastic trend in the sense of cointegration.

To show this, the article splits the portfolio into eigen-trends, trend followers on these principal components. In a Gaussian benchmark, the P&L over a horizon is a quadratic form of returns, and its cumulants follow from a classical formula (Magnus, 1978; Mathai and Provost, 1992). With the same signal in every market, portfolio skewness then has two factors: the single-market skewness (Martin, 2023; Sepp and Lucic, 2026), set by the signal, and a ratio of traces of the eigenvalues of the correlation matrix, its spectrum, set by the correlations and independent of the horizon. This gives one effective number of independent bets, or breadth, for each moment. With equal risk, the Sharpe breadth equals the square of the scaling factor that Valeyre (2025) proposes as an extension of the equicorrelated result of Grebenkov and Serror (2015); the convexity and tail breadths apply known measures of the dispersion of a spectrum to trend P&L. A decomposition of skewness into position and co-movement channels extends the analysis beyond the Gaussian case.

The first finding is that diversification multiplies the bets on the Sharpe ratio but not those on convexity. For 81 markets from 2001 to 2026, the Sharpe breadth is 18.3 but the convexity breadth is 4.1 (Table 2). At this Sharpe breadth, up to 18 convexity bets would be possible; the futures spectrum holds four because a few common trends dominate it. The gap persists in weekly returns (14.3 and 3.3) and without 2008. In the full sample, skewness stops falling beyond about 12 markets while the Sharpe ratio keeps rising (Figure 1).

The second finding is that these bets sit in a few common trends, which also earn the crisis payoff. The first eigen-trend trades a portfolio long equities, commodities, and currencies

and short government bonds and short-rate futures. In the spectrum it carries 51% of the variance but 75% of the convexity (Figure 5). With the next two, it earns 7.7 of the 8.2 percentage points a quarter that the equal-risk portfolio gains gross, before costs, in the 11 worst equity quarters (7.0 with eigenvectors estimated only from earlier years). In the 14 equity bear episodes since 2001, the portfolio gains in 12, on average 11.1% at 10% volatility, while equities lose 25% (Table 3). The concentration rises in stress: in the quarter of days with the highest equity volatility, the convexity breadth is 2.9, against 4.6 on other days (Table 4). Because additional markets mostly load on the same few trends, convexity cannot be bought with breadth.

The third finding is that portfolio design trades these properties against each other. A cross-sectional filter is the matrix that maps the trend signals of all markets into positions. In the benchmark, its eigenvalues, the spectrum, set its convexity; at a given spectrum, its eigenvectors, the directions, set the crisis payoff, because this payoff is tied to the risk-on/risk-off trend. Eigen-trends select directions for variance alone, whereas a trend follower is paid for the trend premium, the expected profit of its positions. Two methods from asset pricing, adapted here to eigen-trends, bring the premium into the choice of directions. Risk-premium principal component analysis (RP-PCA) (Lettau and Pelger, 2020) changes the matrix that is decomposed: it adds the premium matrix of trend P&L to the second moments. Supervised principal component analysis (SPCA) (Giglio, Xiu, and Zhang, 2025) changes the sample that is decomposed: it selects assets by their link to a target factor; the adaptation selects markets by their contribution to the trend premium.

Each design takes a distinct position on the trade-off among skewness, Sharpe ratio, and crisis payoff. In walk-forward tests from 2004, SPCA holds the most skewness, 2.28 gross against 1.49 for the equal-risk portfolio. The estimated optimum, the Sharpe-optimal filter computed from estimated premia, has the highest Sharpe ratio: it gains 0.66 (90% interval 0.37 to 0.97) of gross Sharpe ratio in annual walk-forward tests from 2004 to 2026, but the gain is not confirmed in 1984–2000, falls to 0.23 (-0.08 to 0.52) net of costs with turnover

control, which moves positions 10% of the way to their target each day, and comes with a crisis payoff 6.5 percentage points a quarter lower (−10.6 to −2.8) (Table 6, Figure 7). RP-PCA balances Sharpe ratio and crisis payoff best net of costs: it nearly matches the net Sharpe ratio of the optimum, 0.83 against 0.85, and keeps 7.4 of the 8.1 percentage points of net crisis payoff, at the lowest net skewness of the premium-based filters, with a net Sharpe difference to the equal-risk portfolio of 0.20 (−0.02 to 0.44). Net of costs, the designs thus differ in which of these properties they keep, not in a significant Sharpe gain.

The fourth finding is that the risk rule, which scales the whole portfolio toward a volatility target, decides how much convexity is measured. Convexity runs through the conditional variance of the positions, so holding this variance constant removes the part of the third cumulant generated by unexpected P&L, which formalizes the argument of Hoffman and Kaminski (2016). A rule that scales the current positions by their estimated covariance, traded daily, lowers the 55-day skewness of the equal-risk portfolio traded in instrument positions from 1.38 to 0.07 (Exhibit A20). Traded slowly, the same rule raises the net Sharpe ratio of the reference strategy LS(250,20) of Sepp and Lucic (2026) by 0.10 (0.04 to 0.17) over a matched rule on realized volatility, with a Romano–Wolf adjusted p-value of 0.04 over all 18 comparisons of rule, speed, and comparator (Figure 8; Section D.2.5 of the Appendix). It keeps the crisis payoff of the matched rule in the 14 bear episodes and adds 20 basis points a year (6 to 34) to the certainty equivalent of a 20% trend sleeve on a 60/40 portfolio, 16 after fees (Table 3, Figure 9; Sections E.8 and E.8.2 of the Appendix).

The article contributes to three literatures. For trend following (Moskowitz et al., 2012; Hurst et al., 2017; Martin, 2023; Sepp and Lucic, 2026), it turns two qualitative statements into measurements: that there are few big trends, which is why weight diversification reduces the expected gain of a time-series trend follower (Jusselin, Lezmi, Malongo, Masselin, Roncalli, and Dao, 2017), and that the common trend component preserves skewness under aggregation (Sepp and Lucic, 2026, Section 7.4). It builds on the identity that trading a signal on markets equals trading it on any orthonormal basis of portfolios (Kelly, Malamud,

and Pedersen, 2023, eq. (34)), on the eigen-mode reading of trend portfolios (Benichou, Lempérière, Sérié, Kockelkoren, Seager, Bouchaud, and Potters, 2017; Valeyre, 2023), and on the common trend factor of futures (Chevalier and Darolles, 2025). Its theoretical contribution is the horizon-free factorization of trend P&L into eigen-trends, one count of bets for each moment, and the separation of convexity, set by the spectrum, from crisis payoff, set at a given spectrum by the directions; its empirical contribution is the measurement of these counts on futures, their narrowing in stress, and the location of the crisis payoff in the risk-on/risk-off trend. For portfolio construction from signals and estimated premia (Brandt and Santa-Clara, 2006; Lettau and Pelger, 2020; Kelly et al., 2023; Giglio et al., 2025), it adapts RP-PCA and SPCA to eigen-trends and shows that, with the plug-in optimum, they span the trade-off among skewness, Sharpe ratio, and crisis payoff. For volatility targeting (Moreira and Muir, 2017; Harvey, Hoyle, Korgaonkar, Rattray, Sargaison, and Van Hemert, 2018; Hoffman and Kaminski, 2016; Masturzo, 2025), it splits the effect of a risk rule into channels and finds, at matched trading speed and net of costs, that slow ex-ante targeting raises the net Sharpe ratio of the reference strategy, complementing the correlation-aware scaling of Baltas and Kosowski (2020). The proofs and further evidence are in the online supplement.

2. Data, portfolios, and the eigen-trend decomposition

The research question starts from an empirical regularity: diversification raises the Sharpe ratio of a trend-following portfolio far faster than it dilutes its skewness. The regularity is documented on the futures panel in Section 2.4, and the eigen-trend decomposition of Section 2.5 explains it and organizes the three parts of the answer.

2.1. Data and market universe

The data are a Bloomberg panel of 115 futures markets in seven asset classes, daily from 2 July 1959 to 15 October 2026. Each market is the generic first contract, ratio-adjusted on

Bloomberg's default roll and converted to US dollars as in Sepp and Lucic (2026); returns therefore contain no roll gaps and no collateral return (Carver, 2023). The main tests use the 81 markets with data before July 1999 (equities 16, government bonds 13, short-term interest rates 2, currencies 8, energy 7, metals 10, agriculture 25), evaluated from January 2001 so that every signal and volatility estimate has a full history; the period 1984–2000, on the 21 markets with returns before 1983 (agriculture, metals, and bonds, without equity, currency, or energy futures), is a second sample. Signals, volatility estimates, and portfolio weights use only prior information; the data cleaning of Section G.1 of the Appendix uses the full series. Transaction costs are one-way costs in percent of the notional traded, by asset class and period, from Hurst et al. (2017, Exhibit B1), as in Sepp and Lucic (2026, Table 7.2). Fifteen markets have stale settlement prices and are dropped in the friction checks; because the screen uses full-sample autocorrelations, it is not point in time (Section F.1 of the Appendix). Section G of the Appendix gives the contracts, the roll and cleaning rules, the costs, and a replication inventory.

2.2. Trend-following portfolio and benchmark assumptions

A trend-following strategy is the complete mapping from market returns to time-series signals, instrument positions, cross-sectional portfolio weights, and portfolio-level risk scaling. Because the signal of each portfolio is fixed throughout, the analysis varies only the two portfolio-level choices: the cross-sectional allocation rule, here called the *filter*, and the portfolio risk rule. A filter is a matrix that maps the vector of trend signals into positions, not the time-series signal itself.

There are N markets with daily normalized returns $z_t = (z_{1,t}, \ldots, z_{N,t})^\top$, returns divided by a point-in-time volatility estimate, $z_{i,t} = r_{i,t}/\hat{\sigma}_{i,t-1}$. The filtration $(\mathcal{F}_t)$ satisfies $\sigma(z_s : s \leq t) \subseteq \mathcal{F}_t$ and may contain further information known at t. In every market the trend follower

holds the position

$$S_{t-1} = \sum_{j\geq 1} \beta_j \, z_{t-j}, \qquad \sum_{j\geq 1} \beta_j^2 = 1, \qquad \sum_{j\geq 1} |\beta_j| < \infty, \tag{1}$$

a linear signal of past returns with the same weights β_j in every market, the form common trend indicators share (Levine and Pedersen, 2016). The leading example is the exponentially weighted moving average (EWMA) of Sepp and Lucic (2026), with $\beta_j \propto \nu^{j-1}$ and $\nu = 1 - 2/(\text{span} + 1)$. The signal autocorrelation is $\rho_S(h) = \sum_{j\geq 1} \beta_j \beta_{j+h}$. With positive risk weights $W = \text{diag}(w_1, \ldots, w_N)$, the trend follower in market i earns $f_{i,t} = S_{i,t-1} z_{i,t}$ on day t, and the classic equal-risk portfolio earns

$$g_t = \sum_{i=1}^{N} w_i f_{i,t} = S_{t-1}^\top W z_t, \tag{2}$$

with horizon return $F_T = \sum_{t=1}^{T} g_t$. Convexity is measured by the skewness $\varsigma(F_T) = \kappa_3(F_T)/\kappa_2(F_T)^{3/2}$ of horizon returns, with κ_n the n-th cumulant.

Assumption 1 (Static benchmark)**.** The vectors z_t are independent over time and of $\mathcal{F}_{t-1}$, and Gaussian, $z_t \sim \mathcal{N}(0, P)$, with P a correlation matrix.

Assumption 2 (Conditional model)**.** For every t: (i) $\mathbb{E}[z_t \mid \mathcal{F}_{t-1}] = m_t$; (ii) the innovation $\zeta_t = z_t - m_t$ has the conditional covariance matrix $\mathbb{E}[\zeta_t \zeta_t^\top \mid \mathcal{F}_{t-1}] = \Sigma_t$; (iii) its conditional third moments vanish, $\mathbb{E}[\zeta_{i,t}\zeta_{k,t}\zeta_{l,t} \mid \mathcal{F}_{t-1}] = 0$ for all i, k, l; and (iv) $\sup_t \mathbb{E}\|z_t\|^6 < \infty$.

Assumption 3 (Weak predictability)**.** The process (z_t) is centered, stationary, and Gaussian, with $\mathbb{E}[z_t z_t^\top] = P \succ 0$ a correlation matrix and lag-h autocovariances $\mathbb{E}[z_t z_{t-h}^\top] = \varepsilon A_h$, $h \geq 1$, with $\sum_h \|A_h\| < \infty$ and ε small enough for these autocovariances to define a valid spectral density.

Gaussian or white-noise returns are the standard framework for the analytics of trend-following P&L (Potters and Bouchaud, 2006; Grebenkov and Serror, 2014; Firoozye and

Koshiyama, 2020; Valeyre, 2023; Martin, 2023; Sepp and Lucic, 2026): because the trend follower's convexity arises from the dynamics of its positions rather than from skewed returns, a benchmark without skewed returns and without predictability isolates the mechanism and gives it in closed form. Assumption 1 extends the white-noise benchmark of Sepp and Lucic (2026) to correlated markets: no drift, no autocorrelation, constant correlations, and hence no expected trend-following return, so that it isolates the mechanical effect of diversification. Assumption 2 restores what the static benchmark removes: a predictable drift m_t, the trends from which trend followers earn, and variances and correlations Σ_t that depend on the history, such as the state of the trends. Condition (iii) still rules out skewed daily innovations, so all convexity of the portfolio comes from its dynamics. Assumption 3 adds small serial dependence to the static benchmark; it yields the trend premium and the first-order Sharpe ratio of Section 2.5.

2.3. Signals and portfolios

The tests use two kinds of portfolio, scale-free and implementable. Table A49 in Section G.2.3 of the Appendix defines the variants of EQ and LS(250,20), and Table A1 in Section A defines the filters. Three variants of the classic equal-risk portfolio (EQ) and one reference strategy recur: the scale-free EQ for the tests of the theory, the implementable EQ without turnover control for the mechanism of the risk rules, the controlled EQ, whose positions move each day a share $\eta = 0.1$ of the way to their target (Section 4.6), as benchmark of the net comparisons, and the reference strategy LS(250,20). Most tests of the spectrum use EQ in scale-free form, which trades the normalized trend P&L rather than instrument positions: the equal-risk portfolio $g_t = \sum_i w_i f_{i,t}$ with $w_i = 1/N$ of Equation (2), gross of costs. The tests of the risk rule and of costs instead use implementable portfolios, which trade instrument positions $H_{i,t} = \sigma^* S_{i,t}/\hat{\sigma}_{i,t}$ per unit of capital, with σ^* the instrument volatility target and H_t their vector, net of costs $\sum_i \mathrm{tc}_i |H_{i,t} - H_{i,t-1}|$ at the one-way cost rate tc_i of market i; the signal is an EWMA with a span of 100 days on returns normalized

by an EWMA volatility with a span of 33 days. The reference strategy LS(250,20) of Sepp and Lucic (2026), a signal on the difference of EWMAs with spans of 250 and 20 days, is published and not tuned on these data; in the implementation used here its signal is capped at ±3. Net results trade at the settlement price that forms the signal and exclude roll costs. Two sets of frictions relax these assumptions: the *trading frictions* hold positions one more day between signal and trade, lag cross-market terms, and drop the stale markets, and the *base-case frictions* hold positions one more day, drop the stale markets, and charge roll costs.

Throughout, convexity is measured by the sample skewness of overlapping T-day sums of daily P&L at $T = 55$ days, near the peak of the single-market skewness of a 100-day signal (Martin, 2023; Sepp and Lucic, 2026), and of compounded calendar-quarter returns. Crisis payoff is the mean return in the 11 worst of the 103 S&P 500 quarters, the worst decile (for EQ 8.9% a quarter net without turnover control and 8.1% net with it; the sum of its daily gross P&L in these quarters is 8.2%), and the response to large equity moves is the smile coefficient c of the regression of quarterly net returns on the quarterly return of the S&P 500 futures and its square, $r_q = c_0 + c_1 r_{\mathrm{ES},q} + c\, r_{\mathrm{ES},q}^2 + \upsilon_q$, at 10% volatility; both are reported alongside convexity, not as definitions. The spectral law of Proposition 1 predicts a portfolio's skewness as the mean single-market skewness of its members times the spectral factor of their sample correlation matrix. The channels of Section 3.3 are sample covariances of lagged P&L with the squared P&L and its parts. Because exchanges settle at different times, daily correlations are biased down, and in filters with cross-market terms a later-settling market's signal enters an earlier-settling market's position one day later; weekly returns avoid this bias (Sections C.6 and G.2.4 of the Appendix). Sharpe ratios are annualized with 260 trading days.

All intervals are 90% bootstrap intervals: stationary-bootstrap intervals for Sharpe-ratio differences and moving-block intervals for skewness, which are widened to the 1st to 99th percentiles because nominal skewness intervals cover too rarely. A loss of skewness counts as excluded when the lower bound of the interval of the difference lies above -0.5. Conclusions

rest on rankings across portfolios and on differences rather than on levels (Section 6.3).

2.4. Diversification and convexity

Diversification dilutes convexity far more slowly than it raises the Sharpe ratio (Figure 1). Across 150 random subsets for each number of markets (the 81 single markets for $N = 1$), the median 55-day skewness of the scale-free EQ falls from 2.20 for a single market, close to the white-noise closed form of 2.35, to about 1.25 from 12 markets on, a level far above that of 81 independent markets, $2.20/\sqrt{81} \approx 0.24$, whereas the median Sharpe ratio rises from 0.23 to 0.99. The reference strategy LS(250,20) shows a similar flattening net of costs, from 1.41 for one market to 0.66 at 12 markets (Section C.6 of the Appendix). Because neither the number of markets nor the variance-based effective number of bets accounts for this gap, explaining it requires a count of bets for each moment, which the eigen-trend decomposition provides.

2.5. The eigen-trend decomposition

The decomposition starts from a simple observation. Because a diversified trend follower holds the same signal in every market, it holds that signal on every portfolio of markets, including the principal components of the risk-weighted correlation matrix $M = W^{1/2} P W^{1/2} = \sum_{k=1}^{N} \mu_k q_k q_k^\top$, with eigenvalues $\mu_1 \geq \cdots \geq \mu_N \geq 0$ and orthonormal eigenvectors q_k. The *eigen-trend* k is the trend follower on the eigen-portfolio $y_{k,t} = q_k^\top W^{1/2} z_t$, with daily P&L $\Phi_{k,t} = (q_k^\top W^{1/2} S_{t-1})\, y_{k,t}$; it is a trend follower on one common market factor, and the leading eigen-trend follows the factor with the largest eigenvalue. Under Assumption 1, the eigen-trends are independent trend followers on white noise, each scaled by its eigenvalue (Lemma B.1 in Section B of the Appendix), so the moments of the portfolio depend on the correlations only through the spectrum, the set of eigenvalues. That a signal traded on markets equals the sum of the same signal traded on any orthonormal basis of portfolios is noted by Kelly et al. (2023, eq. (34)), Valeyre (2023) reads the naive Markowitz portfolio as

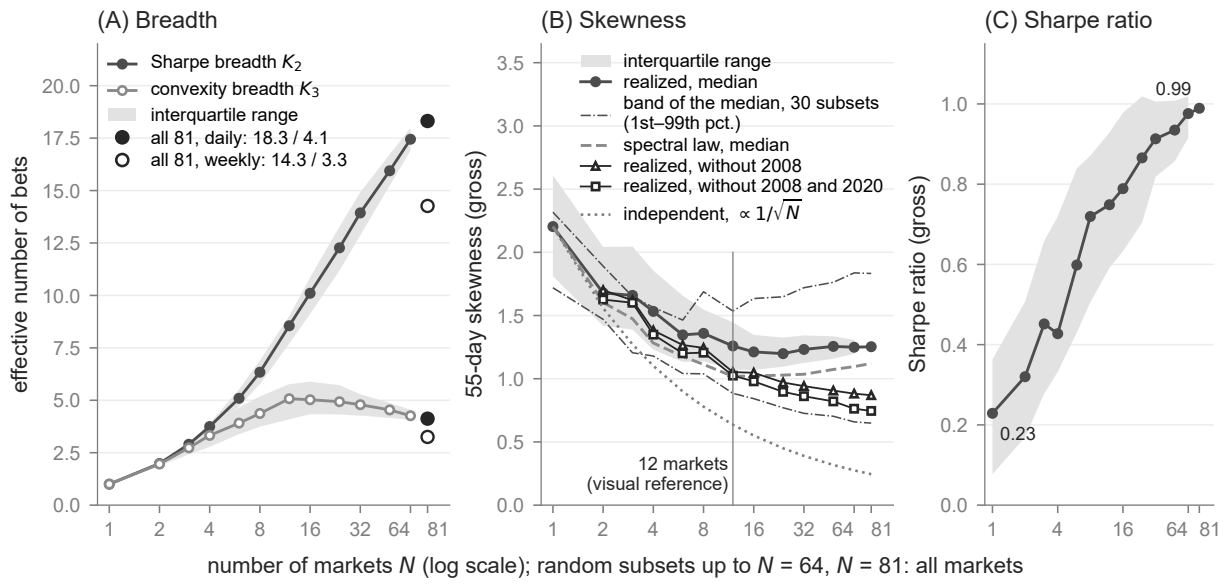


Exhibit 1: **Breadth, Skewness, and Sharpe Ratio against the Number of Markets.** Adding markets keeps raising the Sharpe breadth and the Sharpe ratio, whereas the convexity breadth and skewness flatten beyond about 12 markets in the full sample. Daily returns of 81 futures markets, January 2001–October 2026; scale-free EQ (the equal-risk portfolio in units of normalized returns), gross; 150 random subsets for each N up to 64, all markets at $N = 81$. Panel A: Sharpe breadth K_2 and convexity breadth K_3, effective numbers of independent bets (Corollary 1; median, interquartile range); markers: full panel, correlation from daily (filled) and weekly (hollow) returns. Panel B: 55-day skewness (median and interquartile range of the 150 subsets; widened 90% band of the median from 30 subsets), without 2008 (triangles) and without 2008 and 2020 (squares), the spectral law, and independent markets ($\propto 1/\sqrt{N}$). Panel C: annualized Sharpe ratio, median and interquartile range.

independent trend followers on the eigenvectors and states that the risk of the equal-weight portfolio is in theory proportional to the square root of the eigenvalue, whereas the eigen-trend P&L of Lemma B.1 scales with the eigenvalue, and Jusselin et al. (2017, Appendices A.3.3 and A.5.4) diagonalize the instantaneous P&L in the eigenbasis of its kernel, which with equal exposures and speeds is that of the covariance matrix. New here is the factorization of all cumulants of the horizon return that this independence implies. Benichou et al. (2017) show the risk of portfolio constructions by eigen-mode and name non-quadratic risk measures such as skewness as a route for future work, to identify fundamental risk factors; Proposition 1 gives, separately, the skewness of trend P&L by eigen-mode in the benchmark.

Proposition 1 (Spectral law)**.** *Under Assumption 1, let* Ψ *denote the* T*-day P&L of a single-market trend follower on Gaussian white noise, independent* $\mathcal{N}(0,1)$ *returns (Claim*

B.1 in Section B of the Appendix). Then F_T has the same distribution as $\sum_k \mu_k \Psi_k$ with Ψ_k independent copies of Ψ, and for every $n \geq 2$

$$\kappa_n(F_T) = \kappa_n(\Psi) \sum_{k=1}^{N} \mu_k^n = \kappa_n(\Psi)\, \mathrm{tr}\big((WP)^n\big). \tag{3}$$

In particular, $\mathrm{Var}[F_T] = T\,\mathrm{tr}((WP)^2)$, $\kappa_3(F_T) = 6\sum_{h=1}^{T-1}(T-h)\,\beta_h \rho_S(h)\,\mathrm{tr}((WP)^3)$, *and the skewness and excess kurtosis of the portfolio's T-day return are*

$$\varsigma(F_T) = \varsigma_1(T) \cdot \frac{\mathrm{tr}\big((WP)^3\big)}{\mathrm{tr}\big((WP)^2\big)^{3/2}}, \qquad \delta(F_T) = \delta_1(T) \cdot \frac{\mathrm{tr}\big((WP)^4\big)}{\mathrm{tr}\big((WP)^2\big)^2}, \tag{4}$$

with $\varsigma_1(T)$ and $\delta_1(T)$ the skewness and excess kurtosis of a single market (Grebenkov and Serror, 2014; Martin and Zou, 2012; Martin, 2023), the skewness in the form of Sepp and Lucic (2026, Proposition D.2); the ratio of traces in $\varsigma(F_T)$ is called the spectral factor.

The proposition separates the two ingredients of convexity: the signal and the horizon set the single-market skewness $\varsigma_1(T)$, while the correlations and risk weights scale it by the same spectral factor at every horizon. The factor never exceeds one and equals one only if one common trend drives all markets, a spectral version of the result that diversification lowers skewness (Simkowitz and Beedles, 1978). The proof is short, because F_T is a Gaussian quadratic form with kernel $\mathcal{B}_s \otimes W$, the Kronecker product of the symmetrized time matrix $\mathcal{B}_s$ of the signal weights and the risk weights, so the classical cumulants $\kappa_n = 2^{n-1}(n-1)!\,\mathrm{tr}((\mathcal{Q}\Omega)^n)$ of a quadratic form with matrix $\mathcal{Q}$ in a Gaussian vector with covariance Ω (Magnus, 1978; Mathai and Provost, 1992) factor into $\kappa_n(\Psi)\,\mathrm{tr}((WP)^n)$. The same cumulant formula gives the moments of the instantaneous P&L in the multi-asset trend model of Jusselin et al. (2017), who show numerically that skewness falls with the number of uncorrelated assets, and trace representations exist for a single trend follower (Grebenkov and Serror, 2014) and for linear strategies (Firoozye and Koshiyama, 2020). The contribution of Proposition 1 is the horizon-free factorization of horizon returns, which isolates the cross-section in one num-

ber per moment and in eigen-trends; the corollary below turns each number into an effective count of independent bets.

Corollary 1 (Breadths)**.** *Define for a nonnegative vector $\theta \neq 0$ the Sharpe breadth $K_2(\theta)$, the convexity breadth $K_3(\theta)$, and the tail breadth $K_4(\theta)$, and write $K_n = K_n(\mu)$,*

$$K_2(\theta) = \frac{(\sum_k \theta_k)^2}{\sum_k \theta_k^2}, \qquad K_3(\theta) = \frac{(\sum_k \theta_k^2)^3}{(\sum_k \theta_k^3)^2}, \qquad K_4(\theta) = \frac{(\sum_k \theta_k^2)^2}{\sum_k \theta_k^4}. \tag{5}$$

Under Assumption 1, the skewness and excess kurtosis of the portfolio are exactly $\varsigma_1(T)/\sqrt{K_3}$ and $\delta_1(T)/K_4$; under Assumption 3 instead, K_2 sets the first-order Sharpe ratio (Corollary B.2 in Section B of the Appendix). Moreover,

$$1 \;\leq\; K_4 \;\leq\; K_3 \;\leq\; K_2 \;\leq\; \operatorname{rank}(P) \;\leq\; N, \tag{6}$$

where $K_4 = K_3 = K_2 = \operatorname{rank}(P)$ if and only if all nonzero eigenvalues μ_k are equal, and $K_4 = 1$ if and only if P has rank one.

For the mean, the Sharpe breadth plays the role that K_3 plays for skewness. Under weak predictability (Assumption 3), the Sharpe ratio of the T-day return equals, up to $O(\sqrt{T}\varepsilon^2)$, a leading term called the first-order Sharpe ratio (Proposition B.2(b) in Section B of the Appendix), and K_2 sets it: to first order, the Sharpe ratio of the classic portfolio is the risk-weighted mean single-market Sharpe ratio times $\sqrt{K_2}$, whatever the dispersion of the premia (Corollary B.2 in Section B of the Appendix).

Higher moments count fewer bets because higher powers weight the leading eigenvalues, the common trends, more heavily: if one eigenvalue equals the sum of 80 equal others, $K_2 \approx 4$ but $K_3 \approx 1$. With the variance shares $s_k^{(2)} = \mu_k^2 / \sum_j \mu_j^2$ of the eigen-trends, K_2, K_3, and K_4 are the exponentials of the Rényi entropies of $s^{(2)}$ of orders $\frac{1}{2}$, $\frac{3}{2}$, and 2, so the ordering (6) is the monotonicity of Rényi entropy in its order (Rényi, 1961); the breadths are Hill numbers of the variance shares (Hill, 1973), and their Hill number of order one, the effective number

of bets of Meucci (2009) applied to the eigen-trends as uncorrelated bets, lies between K_3 and K_2.

The breadths also coincide with several established measures. K_2 is the participation ratio of the spectrum and the Satterthwaite degrees of freedom (Satterthwaite, 1946; Box, 1954); it extends the breadth of Grinold (1989) to correlated bets and, with equal risk, equals $N/(1+(N-1)\overline{\rho^2})$, with $\overline{\rho^2}$ the mean squared pairwise correlation, the scaling that Valeyre (2025) proposes as an extension of the equicorrelated result of Grebenkov and Serror (2015) and tests on data. K_3 is the degrees of freedom of the three-moment χ^2 approximation (Pearson, 1959; Imhof, 1961), and K_3 and K_4 are the dimensionalities of Barkhagen, Fleming, García Quiles, Gondzio, Kalcsics, Kroeske, Sabanis, and Staal (2023) when the reference asset is the single-market trend follower. New is the identification of the moments of diversified trend P&L with Hill numbers of the eigen-trend variance shares, which is horizon-free and the same for every linear signal; the ordering then follows from the monotonicity of Hill numbers. The breadths are statistics of the return correlation matrix; specific to trend following is that they govern the moments of trend P&L, a property that the realized-implied counts, reported as diagnostics, test.

The first prediction, P1, follows: for a given signal, the spectrum sets the convexity of a diversified portfolio in the benchmark and should order it in the data, with skewness falling with K_3 rather than with the number of markets while the Sharpe ratio rises with K_2.

3. Sharpe breadth and convexity breadth

The eigen-trend decomposition turns the first part of the research question, how many independent bets diversification creates for the Sharpe ratio and for convexity, into a measurement of the spectrum. The measurement shows a large gap between the two breadths, robust to the crises, and locates the convexity, together with the crisis payoff, in the leading eigen-trend.

3.1. Breadth of the futures universe

A large futures universe offers many more effective bets on the Sharpe ratio than on convexity. The full panel has a Sharpe breadth of 18.3, a convexity breadth of 4.1, and a tail breadth of 3.3 from daily returns, and 14.3, 3.3, and 2.7 from weekly returns (Table 2). Beyond 12 markets, additional markets load on the common trends, and K_3 falls slightly, from its peak of 5.1 to 4.1 (Panel A of Figure 1). With about four convexity bets, portfolio skewness is about half of single-market skewness, the ratio $1/\sqrt{4}$ of four independent trends. The Sharpe breadth, in turn, gives the order of magnitude of the realized Sharpe ratio: by Corollary B.2, the mean full-sample single-market Sharpe ratio of 0.274 (Section C.1 of the Appendix) implies $0.274\sqrt{18.3} = 1.17$ against a realized 0.99, a shortfall consistent with the excess co-movement of trend P&L (Section 3.3) and with asynchronous closes (1.04 from weekly breadth) that, to first order, cannot come from dispersion of the premia.

Two checks rule out mechanical explanations of the gap. With 81 markets and about 6,700 trading days, noise in the sample spectrum is negligible: Gaussian panels drawn with the sample correlation as the true matrix reproduce the three breadths within 0.4% (Section C.6 of the Appendix), so the relevant uncertainty comes from non-stationarity, asynchronous closes, and fat tails, which the bootstrap bands, the weekly row, and the rolling windows address. The gap is also not the consequence of one dominant factor alone. At $N = 81$ and $K_2 = 18.3$, every nonnegative spectrum has a convexity breadth between 1.6, that of an equicorrelated matrix, and 18.1; the futures spectrum, at 4.1, holds more convexity bets than a one-factor structure but lies in the lowest sixth of this range (reference rows of Table 2).

The spectrum, not the number of markets, accounts for skewness. Across 34 cells of portfolio design and size, K_3 accounts for 81% of the variation of the ratio of portfolio to member skewness against 64% for the number of markets, and in a joint regression the coefficient on $1/\sqrt{N}$ is indistinguishable from zero, while that on $1/\sqrt{K_3}$ is 0.51 with an intercept of 0.28. The spectrum thus accounts for most of the variation across portfolios,

Universe	Markets	Spectral breadth				Skewness, 55-day		Sharpe ratio
		K_2	K_3	K_4	K_3/K_2	Realized	Spectral law	
Full universe, daily returns	81	18.3	4.1	3.3	0.23	1.25	1.12	0.99
		[16.0, 19.6]	[3.2, 4.8]	[2.6, 3.8]				
Full universe, weekly returns	81	14.3	3.3	2.7	0.23	1.25	1.26	0.99
		[12.3, 15.2]	[2.6, 3.8]	[2.2, 3.2]				
Full universe, without 2008	81	18.7	4.3	3.4	0.23	0.87	1.11	0.88
Full universe, without 2008 and 2020	81	19.0	4.5	3.6	0.23	0.75	1.07	0.86
Full universe, equal risk per asset class	81	14.6	4.3	3.7	0.30	1.42	1.10	0.99
		[13.1, 15.3]	[3.6, 4.8]	[3.0, 4.1]				
1984–2000	21	9.5	3.6	2.8	0.37	1.04	1.19	0.99
Reference: equicorrelated, same K_2 ($\rho = 0.21$)	81	18.3	1.6	1.4	0.08	–	–	–
Reference: class-block average correlations	81	23.3	3.7	3.0	0.16	–	–	–

Exhibit 2: **Spectral Breadth and Skewness by Universe and Sample.** Scale-free P&L of a single EWMA signal (span 100), equal risk, gross; full universe: 81 markets, January 2001–October 2026. K_2, K_3, K_4: Sharpe, convexity, and tail breadth (Corollary 1) of the correlation matrix of daily or non-overlapping 5-day returns, 90% moving-block intervals below. Realized: skewness of overlapping 55-day sums; spectral law: mean single-market skewness times the spectral factor (Proposition 1). Without 2008 (and 2020): windows touching these years dropped. Equal risk per asset class: one seventh of the risk per class. 1984–2000: the 21 markets with earlier data. Reference rows: equicorrelated matrix with the daily K_2, and class-average correlations; –: not applicable.

although level and slope differ from the static benchmark; market selection that raises K_3 in sample does not keep its skewness out of sample (Section C.1 of the Appendix). The gap between the breadths is general: in rolling ten-year windows, random and class-balanced universes, with equal risk per asset class, and in 1984–2000, the convexity breadth stays below half of the Sharpe breadth (Table 2; Sections C.2 and C.6 of the Appendix).

3.2. Crises and the concentration of convexity

The spectral gap is robust to the crises; the realized dilution of skewness is not. Without 2008, K_2 and K_3 are 18.7 and 4.3, and without 2008 and 2020, 19.0 and 4.5 (Table 2). In the full sample skewness is flat from about 12 markets on: the median at 12 markets differs from the full panel by 0.04 (widened interval −0.39 to 0.35; Section C.6 of the Appendix). Without 2008, skewness keeps falling slowly, from 1.05 at 12 markets to 0.87 at 81, and without 2008 and 2020 to 0.75, still three times the level of independent markets. Translated through the benchmark identities, realized skewness corresponds to about 3 convexity bets in the full sample, 7 without 2008, and 9 without both years, against realized-implied Sharpe breadths of 13 to 16 (Section C.6 of the Appendix). The crisis years change the realized convexity,

not the convexity that the spectrum offers. The first prediction, P1, holds: diversification is moment-specific, the spectrum, more than the number of markets, accounts for convexity, and skewness dilutes slowly beyond a dozen markets, at a level set by the crises in the sample. Few convexity bets imply that convexity is concentrated, and the spectrum shows where.

Convexity sits in the strongest eigen-trends, which hold more of the convexity than of the variance, because higher powers weight the largest eigenvalues more (Corollary B.1 in Section B.2 of the Appendix). The spectrum puts 51% of the variance of the trend P&L and 75% of its convexity in the first eigen-trend (90% bands 45% to 59% and 67% to 83%), and 77% and 94% in the first three (Panel A of Figure 5). The realized P&L confirms this concentration for the leading trends jointly: counting co-skewness, the first three eigen-trends contribute 97% of the realized third moment of 55-day returns against 83% of the variance, and the first alone 57% (90% band 16% to 88%; Section C.3 of the Appendix). Few convexity bets and the concentration of convexity in a few common trends are two readings of the same spectrum.

The leading eigen-trend has an economic reading and carries the crisis payoff. Its eigenvector loads positively on equities, commodities, and currencies and negatively on government bonds and short-term interest-rate futures, a risk-on/risk-off trend; equities hold 46% and bonds 19% of its squared loadings (Section E.8 of the Appendix), unlike the risk-parity portfolio against which Dao et al. (2017) find the convexity of a trend program. An exact split of the trend P&L along the eigenvectors locates the crisis payoff: of the 8.2% a quarter that the controlled EQ earns gross in the 11 worst equity quarters, 5.9 percentage points come from the first eigen-trend, which carries 43% of its position risk, and 7.7 from the first three (Panel D of Figure 5; Section E.8 of the Appendix). The concentration in the first three does not rest on hindsight: with eigenvectors re-estimated each year from earlier data only, the first three still carry 7.0 and the first 3.8, because the leading direction shifts; from 2022 to 2025, when the stock–bond correlation was positive, bonds held little of its squared loadings (Section C.3 of the Appendix). In market terms the two outcomes differ. Every asset class

contributes to convexity roughly in proportion to its risk (agriculture 26% of the variance and 32% of the convexity, equities 19% and 15%, bonds 18% and 14%), whereas bonds (2.8 percentage points) and equities (1.9) earn more than half of the crisis payoff with 37% of the risk, mainly through the timing of their positions (Sections C.3 and E.8 of the Appendix). Convexity is thus a property of a few common directions rather than of particular markets, whereas the crisis payoff runs mainly through the bond–equity pair.

The few bets reflect links across asset classes rather than the number of markets, and they narrow in stress (Table 4). Class-average correlations alone give a convexity breadth of 3.7, whereas without correlation across classes it rises to 7.9. The two leading eigen-trends have an economic reading that holds out of sample, with eigenvectors from one half of the sample and returns of the other: the first is a bet on growth and risk appetite, moving with equities, metals, and energy and against bonds, and the second moves with currencies against the dollar, short rates, and metals. Together they carry 89% of the convexity in the spectrum. The reading of the first also holds against financial conditions: with full-sample eigenvectors, its quarterly P&L rises by 1.6 to 1.7 percentage points with a one-standard-deviation tightening in the credit spread, the VIX, or the Chicago Fed financial conditions index (t-statistics 2.7 to 4.0), whereas the groups of the other eigen-trends do not respond significantly. The response runs through equity sell-offs: with the S&P 500 return as a control, the slopes are 0.6 to 1.0 (t-statistics 0.6 to 1.8). In the top quartile of S&P 500 volatility up to the previous day the breadth is 2.9 against 4.6 on other days (difference -1.7, 90% interval -2.5 to -0.9), and the first eigen-trend holds 86% against 69% of the convexity, so the portfolio is most concentrated when convexity matters most. The narrowing does not depend on the stress measure: in the top quartile of each of these three series the breadth is 2.4 to 2.5, against 5.0 to 5.1 on other days (Section C.3 of the Appendix). In every sub-period the first four ranks hold at least 95% of the convexity, at a breadth between 2.3 and 5.0 (Section C.3 of the Appendix).

Equity bear episodes, dated by market turning points rather than calendar quarters,

confirm this concentration (Table 3). The 14 bear episodes of the S&P 500 futures since 2001 are falls of more than 15% from peak to trough, each confirmed once the futures have recovered 15% from the trough, the symmetric (15,15) filter of Lunde and Timmermann (2004). The controlled EQ gains in 12 of them, on average 11.1% at 10% volatility while equities lose 25%, and the first eigen-trend carries 7.0 percentage points of its gross P&L in these episodes (5.0 with real-time eigenvectors; Section C.3 of the Appendix) and is positive in 11 episodes. The rate shock of 2022 is the exception: equities and bonds fell together, the risk-on/risk-off trend earned little, and the payoff of 30 percentage points came from the trends in rates and commodities carried by the other eigen-trends. The largest payoffs come in persistent sell-offs (2001–2002, 2007–2008, 2022), and the payoff is small or negative in the short reversals of 2010 and 2025. The gap to the estimated optimum does not depend on the dating rule: with recovery thresholds of 10% or 20%, EQ gains 8.3% and 16.2% on average, against 1.5% and 4.0% for the estimated optimum of Section 4 (Section E.8.1 of the Appendix).

			Classic EQ		Filters					Risk rule				
Episode	Peak–trough	S&P 500	Total	Eigen-trend 1	MKT	RP-PCA	SPCA	OPT	C_ϑ	EQ, ex post	EQ, ex ante	LS	LS, ex post	LS, ex ante
Dot-com bust	01/2001–04/2001	−16	4.0	5.7	−0.3	2.2	0.3	−1.1	0.2	4.8	5.7	4.6	5.8	4.7
Dot-com bust	05/2001–09/2001	−27	18.8	16.1	7.7	18.1	19.1	6.4	10.1	21.1	21.5	19.4	20.7	20.5
Dot-com bust	01/2002–07/2002	−32	12.0	2.1	7.5	5.5	5.5	−5.3	−4.0	15.2	11.5	9.2	10.7	11.2
Dot-com bust	08/2002–10/2002	−19	11.8	3.8	10.7	8.9	6.0	5.7	5.7	7.4	7.8	12.0	8.5	6.8
Global financial crisis	10/2007–10/2008	−48	47.3	21.4	34.5	37.2	39.3	15.7	17.1	34.7	39.1	26.4	21.4	25.2
Global financial crisis	11/2008 (4–20 Nov.)	−25	13.7	12.2	12.0	9.0	11.6	3.0	3.0	4.7	4.8	10.0	5.2	4.3
Global financial crisis	01/2009–03/2009	−27	8.3	8.1	7.7	4.2	6.2	5.4	3.8	3.2	5.3	13.3	6.2	5.1
Euro crisis, flash crash	04/2010–07/2010	−16	−2.0	−1.8	−3.6	1.9	−4.8	−3.1	−1.3	−2.8	−1.2	−3.2	−4.4	−3.8
Euro crisis, US downgrade	04/2011–10/2011	−19	5.1	9.9	−0.7	−2.4	−4.2	0.2	1.0	4.4	5.7	0.7	1.8	6.0
Fed tightening	09/2018–12/2018	−20	3.6	6.0	2.5	2.7	5.1	−5.5	−4.2	4.3	3.7	−0.6	−1.5	−1.7
Covid-19	02/2020–03/2020	−34	7.1	12.4	2.3	4.9	2.1	−4.9	−4.1	6.1	−0.2	−4.9	−7.0	−8.9
Inflation and rate shock	01/2022–06/2022	−23	23.0	4.0	33.9	21.0	31.8	16.3	19.1	23.1	24.8	29.0	25.5	26.3
Inflation and rate shock	08/2022–10/2022	−17	7.1	−0.3	4.3	8.9	2.6	6.8	6.3	6.9	9.4	11.9	9.8	11.7
Tariff shock	12/2024–04/2025	−19	−5.0	−1.8	−0.8	−2.2	−4.2	−3.9	−3.4	−8.0	−7.6	−6.0	−10.9	−11.0
Mean		−25	11.1	7.0	8.4	8.6	8.3	2.6	3.5	8.9	9.3	8.7	6.6	6.9
Positive episodes			12 of 14	11 of 14	10 of 14	12 of 14	11 of 14	8 of 14	9 of 14	12 of 14	11 of 14	10 of 14	10 of 14	10 of 14

Exhibit 3: **Trend Payoff in Equity Crisis Episodes.** Bear episodes of the S&P 500 futures, January 2001–October 2026: a fall of more than 15% from a peak, ending at the trough once the futures have recovered 15% (Lunde and Timmermann, 2004). S&P 500: return over the episode (%). Other columns: cumulative net return (%) of the implementable portfolios on 81 markets with turnover control ($\eta = 0.1$) at 10% volatility, no portfolio rule unless marked ex post (realized-volatility rule, span 100) or ex ante, both at $\eta = 0.1$; filters of Table 6; LS: LS(250,20). EQ total: the whole portfolio; eigen-trend 1: its gross part carried by the first eigenvector (Section E.8 of the Appendix). Positive episodes: count with a gain. Alternative dating rules and crisis clusters: Section E.8.1 of the Appendix. Descriptive.

Panel A: Shares of the spectrum and squared loadings by asset class (%)

Rank	Variance	Convexity	Equities	Bonds	STIR	FX	Energy	Metals	Agriculture
1	51	75	46	19	1	6	8	11	9
2	17	14	1	42	3	21	5	18	10
3	9	5	37	14	2	1	10	0	36
4	5	2	10	7	1	11	14	8	48

Panel B: Out-of-sample correlation with the class return, the markets of the class left out of the eigen-portfolio

Rank	Trained	Equities	Bonds	STIR	FX	Energy	Metals	Agriculture
1	2001–2012	0.42	−0.24	−0.27	0.16	0.36	0.35	0.31
	2013–2026	0.57	−0.45	−0.46	0.30	0.33	0.46	0.38
2	2001–2012	0.11	0.12	0.30	0.34	0.02	0.26	0.07
	2013–2026	−0.14	−0.09	0.36	0.36	0.22	0.22	0.15
3	2001–2012	−0.14	0.00	0.29	0.20	−0.06	0.15	0.12
	2013–2026	−0.39	−0.15	0.02	−0.20	−0.22	−0.20	−0.10
4	2001–2012	−0.26	−0.07	0.21	0.09	0.07	0.07	−0.20
	2013–2026	−0.33	−0.20	0.15	0.07	−0.08	0.04	−0.34

Panel C: Breadth and concentration

Correlation matrix	K_2	K_3	Rank 1	Ranks 1–4
All days	18.3	4.1	75%	96%
Top quartile of S&P 500 volatility	15.2	2.9	86%	98%
Other days	19.2	4.6	69%	95%
Class-average correlations	23.3	3.7	78%	98%
No correlation across classes	24.8	7.9	44%	86%
2001–2008	18.8	4.6	68%	95%
2009–2014	12.6	2.3	91%	98%
2015–2020	17.5	3.7	82%	96%
2021–2026	18.6	5.0	55%	96%
Independent markets	79.9	77.9	2%	9%

Exhibit 4: **The Few Convexity Bets: Composition, Reading, and Stability.** 81 markets, January 2001–October 2026, full-sample correlation of normalized returns unless stated. Variance and convexity: shares $\mu_k^2/\sum_j \mu_j^2$ and $\mu_k^3/\sum_j \mu_j^3$ of the spectrum. Panel B: eigenvectors from the sample named, returns of the other half, orientation of the full-sample vector; class return: equal-weight mean of the normalized returns of the class. Panel C: stress: 63-day S&P 500 volatility up to the previous day above its full-sample 75th percentile. Stress minus other days: K_3 **−1.72** [−2.47, −0.88], share of rank 1 **0.16** [0.08, 0.25] (moving-block bootstrap, 63 days, 1,000 draws). Block bootstrap of K_3, 5th to 95th percentile: 3.6 to 4.7. Leaving out one class: 3.2 to 5.1. Random subsets of 20, 40, and 60 markets, median: 5.0, 4.7, 4.3.

3.3. Channels of convexity

The spectral law describes a static benchmark in which correlations are constant and returns unpredictable. In the data, trends exist and correlations move with them (Assumption 2), so convexity can arise through more than one route. To separate the routes, the daily P&L is split into a predictable drift $d_t = \mathbb{E}[g_t \mid \mathcal{F}_{t-1}]$, the expected trend profit, and an innovation $e_t = g_t - d_t$, the unexpected part, with conditional variance

$$V_t = \mathbb{E}[e_t^2 \mid \mathcal{F}_{t-1}] = S_{t-1}^\top W \Sigma_t W S_{t-1}, \tag{7}$$

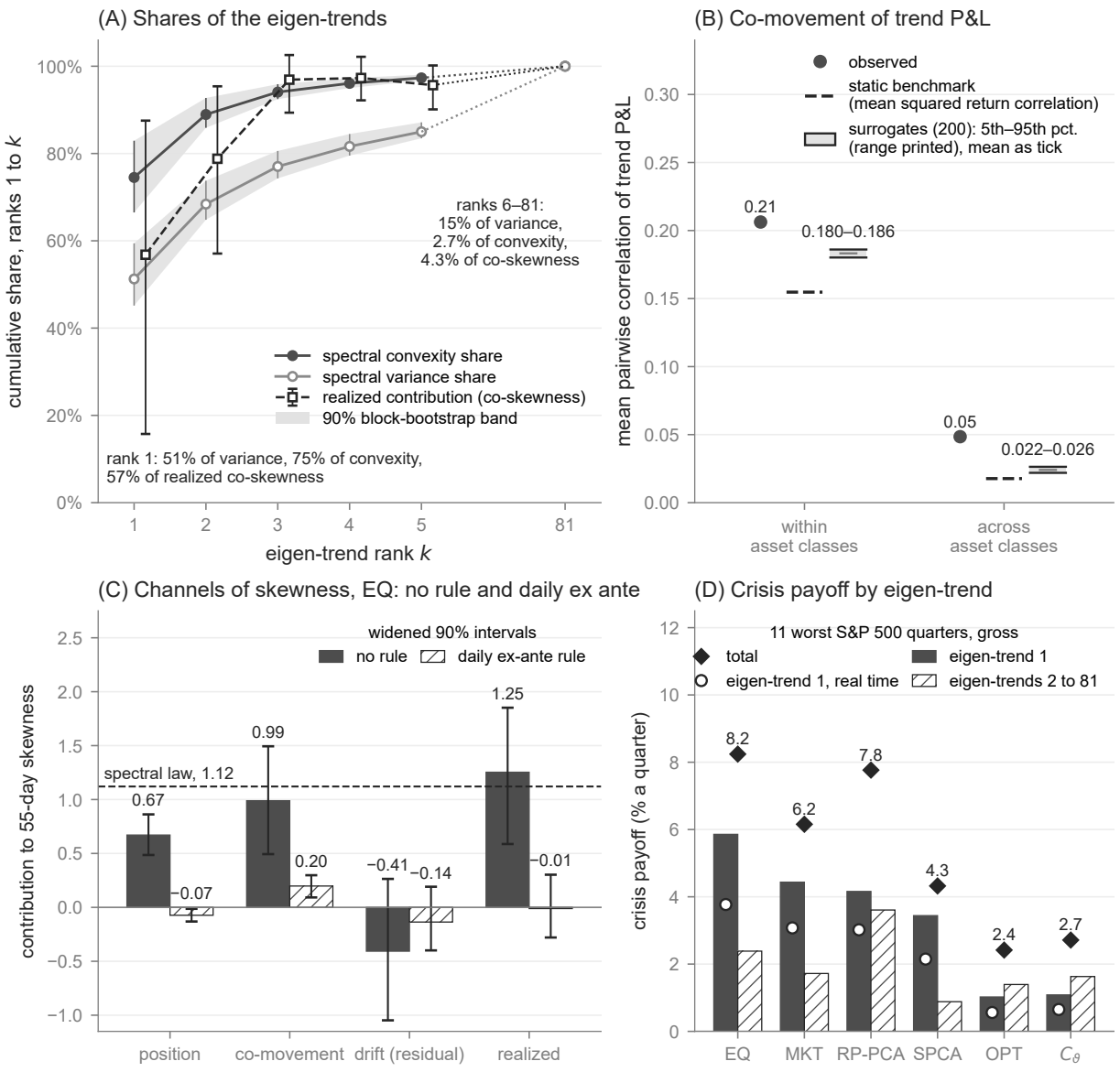


Exhibit 5: **Eigen-Trends, Co-Movement, Channels, and Crisis Payoff.** The leading eigen-trends hold more of the convexity than of the variance and, for EQ, most of the crisis payoff. 81 futures markets, January 2001–October 2026; scale-free EQ, 55-day horizon. Panel A: cumulative variance and third-moment shares of the eigen-trends implied by the spectrum (90% bands) and the realized contribution with co-skewness (dashed, 90% intervals). Panel B: mean pairwise correlation of trend P&L within and across asset classes, observed, static benchmark (the mean squared return correlation), and 200 phase-randomized surrogates (5th–95th percentiles). Panel C: channels of skewness (Proposition 2), no rule (filled) and daily ex-ante rule (hatched), widened 90% intervals; drift: residual; dashed: the spectral law. Panel D: P&L in the 11 worst S&P 500 quarters from eigen-trend 1 (full-sample eigenvectors; hollow: real-time eigenvectors) and the rest, for EQ and the filters of Table 6, diamonds: total; turnover control ($\eta = 0.1$), gross, 10% volatility.

so the horizon return splits into a martingale sum $E_T = \sum_t e_t$ and a drift sum $D_T = \sum_t d_t$. For a fixed matrix $\bar{P}$, such as $\mathbb{E}[\Sigma_t]$, the conditional variance splits into a position part and

parts due to moving variances and correlations,

$$V_t^{\text{pos}} = S_{t-1}^{\top} W \bar{P} W S_{t-1}, \quad V_t^{\text{var}} = \sum_i w_i^2 S_{i,t-1}^2 \big(\Sigma_{t,ii} - \bar{P}_{ii}\big), \quad V_t^{\text{cor}} = \sum_{i \neq l} w_i w_l S_{i,t-1} S_{l,t-1} \big(\Sigma_{t,il} - \bar{P}_{il}\big). \tag{8}$$

Convexity of the martingale part requires that the risk of the portfolio rise after it has gained: through larger positions at fixed correlations, the *position channel* of the benchmark, or through variances and correlations that depend on the past, the *co-movement channel*.

Proposition 2 (Channels of convexity)**.** *Under Assumption 2:*

(a) $\operatorname{Var}[E_T] = \sum_{t=1}^{T} \mathbb{E}[V_t]$ *and*

$$\kappa_3(E_T) = 3 \sum_{1 \leq s < t \leq T} \operatorname{Cov}(V_t, e_s) = \mathcal{C}_T^{\text{pos}} + \mathcal{C}_T^{\text{var}} + \mathcal{C}_T^{\text{cor}}, \tag{9}$$

with $\mathcal{C}_T^{\bullet} = 3 \sum_{1 \leq s < t \leq T} \operatorname{Cov}(V_t^{\bullet}, e_s)$ *for* $\bullet \in \{\text{pos}, \text{var}, \text{cor}\}$.

(b) $\kappa_3(F_T) = \kappa_3(E_T) + \Delta_T$*, with* $\Delta_T = 3\,\mathbb{E}[E_T^2 \tilde{D}_T] + 3\,\mathbb{E}[E_T \tilde{D}_T^2] + \mathbb{E}[\tilde{D}_T^3]$ *and* $\tilde{D}_T = D_T - \mathbb{E}[D_T]$*;* Δ_T *is the drift part of the third cumulant.*

(c) For the classic portfolio under Assumption 1 and with $\bar{P} = P$*:* $\Delta_T = 0$*,* $\mathcal{C}_T^{\text{var}} = \mathcal{C}_T^{\text{cor}} = 0$*, and* $\mathcal{C}_T^{\text{pos}}$ *is the third cumulant of Proposition 1.*

(d) If (z_t, m_t, Σ_t) *is strictly stationary,* $\kappa_3(E_T) = 3 \sum_{h=1}^{T-1} (T-h) \operatorname{Cov}(V_t, e_{t-h})$*, and likewise for each* $\mathcal{C}_T^{\bullet}$.

Part (a) applies the skewness identity of Neuberger and Payne (2021) to trend P&L, and the split into channels is new. Because the split is an accounting identity that depends on $\bar{P}$, it measures where the third cumulant sits, not what causes it. Part (b) adds the drift part, the contribution of the predictable trend profit, which vanishes in the static benchmark (part (c)).

Under stationarity and with $\bar{P} = \mathbb{E}[\Sigma_t]$, the correlation part $\mathcal{C}_T^{\text{cor}}$ is called the correlation channel; it is positive when the correlations between the positions rise after the portfolio has gained (Definition D.1 in Section D of the Appendix).

A positive correlation channel is a signature of trends that move correlations (Safari and Schmidhuber, 2026), but persistent correlation regimes, the rise of correlations in sell-offs and panics (Longin and Solnik, 2001; Borland, 2012), and the leverage effect on correlations, in stock indices (Reigneron, Allez, and Bouchaud, 2011) and in futures (Karami, Benichou, Benzaquen, and Bouchaud, 2021), produce it as well. The drift part, by contrast, is negative when trends reverse after large moves (Schmidhuber, 2021; Daniel and Moskowitz, 2016; Goulding, Harvey, and Mazzoleni, 2023), so the two non-benchmark terms can offset each other.

In the data the co-movement channel adds more convexity than the position channel, and the residual drift part offsets part of it. For the scale-free EQ, the decomposition is additive in skewness units (Panel C of Figure 5): the realized 1.25 is the position channel, 0.67, plus the co-movement channel, 0.99, plus the drift part, -0.41. Phase-randomized surrogates reproduce the position channel but not the co-movement channel, which comes almost entirely from its correlation part. A stricter null that flips the sign of all returns of a day, in blocks of five days, keeps volatility and correlation clustering but removes trends; it reproduces the size of the correlation part (0.93 against a mean of 0.82, $p = 0.28$), while the excess co-movement of trend P&L within asset classes exceeds every draw (Panel B of Figure 5; Section D.1 of the Appendix). The correlation channel therefore reflects the clustering of risk in crises, of which trends that move correlations are one source. Realized skewness lies close to the spectral law with empirical single-market inputs, 1.12, but not because the non-benchmark terms cancel: the position channel lies 0.45 below the law, and the co-movement channel and the drift part together add 0.58. The drift part is a residual that also collects estimation terms (Section G.2.6 of the Appendix). The correlation part, like the excess of realized skewness over the law, is concentrated in 2008 and 2020 (Sections

C.6 and D.1 of the Appendix).

The answer to the first question is that a diversified trend follower holds many Sharpe bets but few convexity bets, and that its crisis payoff and most of its convexity sit in a few common trends, led by a risk-on/risk-off trend; the co-movement channel adds convexity mainly in crises. Because convexity cannot be bought with breadth, the design question is not how many markets to hold but which cross-sectional directions to trade, how to allocate risk across their spectrum, and how to align those directions with the trend premia.

4. Cross-sectional design

Section 3 shows that the available convexity is concentrated in a few common trends. The second part of the research question is whether cross-sectional design can use this structure to improve the trade-off between the Sharpe ratio and convexity, and how much of an improvement survives implementation. To answer it, the theory separates three properties of a filter, its directions, its spectrum, and its alignment with the premia, and the evidence follows the candidate designs from gross to net performance.

4.1. Trend factors: directions, spectrum, and alignment

A *cross-sectional filter* is a symmetric $N \times N$ matrix C that maps the signals into positions, $x_{t-1} = CS_{t-1}$, so that the portfolio earns $g_t^C = S_{t-1}^\top C z_t$; the classic filter EQ is $C = W$. The analysis is restricted to symmetric filters, the symmetric linear strategies of Kelly et al. (2023); the antisymmetric part, their principal alpha portfolios, trades lead–lag premia, which carry little of the trend premium in this panel (Section E.4.1 of the Appendix), and directed network filters also lie outside this class. Every positive semidefinite filter can be written as $C = \sum_k \omega_k u_k u_k^\top$: it trades the *directions* u_k, portfolios of the normalized returns, as trend followers with weights $\omega_k \geq 0$. Each direction defines a *trend factor*, the trend follower on the portfolio u, with daily return $(u^\top S_{t-1})(u^\top z_t)$, so a filter is a portfolio of trend factors, and the eigen-trends of Section 2.5 are the trend factors of EQ. With the directions

chosen P-orthonormal ($P \succ 0$), the weights ω_k are the eigenvalues of CP, the *spectrum* of the filter, which states how much risk it places on each trend factor, and directions and spectrum separate exactly. The premium side is described by the trend cross-moment and the premium matrix

$$\Gamma = \mathbb{E}\big[z_t S_{t-1}^\top\big], \qquad \Gamma_s = \tfrac{1}{2}\big(\Gamma + \Gamma^\top\big), \qquad \Pi = W^{1/2}\Gamma_s W^{1/2}, \tag{10}$$

a filter earns $\mathbb{E}[g_t^C] = \operatorname{tr}(C\Gamma)$, and a trend factor earns the quadratic form $u^\top \Gamma_s u$; the diagonal of Γ holds the own-market premia $\pi_i = \Gamma_{ii}$ and its off-diagonal terms lead–lag premia (Lemma B.2 in Section B of the Appendix). The premium matrix thus collects the expected trend profit of every pair of markets, and the *alignment* of a filter describes how its directions load on these premia, that is, whether it trades the trend factors that earn most.

Because a filter is a portfolio of trend factors, the spectral law carries over with the eigenvalues ϑ_k of CP in place of μ_k: under Assumption 1 the horizon return $F_T^C = \sum_{t=1}^T g_t^C$ has skewness $\varsigma(F_T^C) = \varsigma_1(T)\operatorname{tr}((CP)^3)/\operatorname{tr}((CP)^2)^{3/2}$, and under Assumption 3 its first-order Sharpe ratio is $\sqrt{T}\operatorname{tr}(C\Gamma)/\operatorname{tr}((CP)^2)^{1/2}$ (Proposition B.2 in Section B of the Appendix).

Proposition 3 (Sharpe ratio at a given spectrum)**.** *Under Assumption 3, let $\Gamma_s \neq 0$, let $\lambda_1 \geq \cdots \geq \lambda_N$ be the eigenvalues of $\tilde{\Gamma} = P^{-1/2}\Gamma_s P^{-1/2} = \sum_k \lambda_k \tilde{q}_k \tilde{q}_k^\top$, $\lambda^+ = (\max(\lambda_k, 0))_k$, and $\|\cdot\|$ the Euclidean norm. Let $C \succeq 0$, $C \neq 0$, with eigenvalues $\vartheta_1 \geq \cdots \geq \vartheta_N \geq 0$ of CP. (a) The first-order Sharpe ratio satisfies $\sqrt{T}\operatorname{tr}(C\Gamma)/\operatorname{tr}((CP)^2)^{1/2} \leq \sqrt{T}\sum_k \vartheta_k\lambda_k/(\sum_k \vartheta_k^2)^{1/2}$, with equality for the filter $C_\vartheta = P^{-1/2}\big(\sum_k \vartheta_k \tilde{q}_k \tilde{q}_k^\top\big)P^{-1/2}$ that has the same spectrum and is aligned with the premium. With returns in the benchmark, that is, at $\varepsilon = 0$, and the filters held fixed, both have the same skewness, $\varsigma_1(T)\sum_k \vartheta_k^3/(\sum_k \vartheta_k^2)^{3/2}$. (b) For $\bar{K} \geq 1$ let $\mathrm{SR}^{\max}(\bar{K})$ be the largest first-order Sharpe ratio of a filter $C \succeq 0$ whose convexity breadth $K_3(\vartheta)$ is at most $\bar{K}$. If $\lambda_1 > 0$, $\mathrm{SR}^{\max}$ is non-decreasing, $\mathrm{SR}^{\max}(\bar{K}) = \sqrt{T}\|\lambda^+\|$ for $\bar{K} \geq K_3(\lambda^+)$, and $\mathrm{SR}^{\max}(\bar{K}) < \sqrt{T}\|\lambda^+\|$ for $\bar{K} < K_3(\lambda^+)$; for $\bar{K} \geq K_3(\lambda^+)$ the maximum is attained by the Sharpe-optimal positive semidefinite filter, up to scale $C^+ = P^{-1/2}(\sum_k \lambda_k^+ \tilde{q}_k \tilde{q}_k^\top)P^{-1/2}$*

(Proposition B.1 in Section B of the Appendix).

The proposition is easiest to read in the whitened directions $P^{-1/2}\tilde{q}_k$, with $\tilde{q}_k$ the eigenvectors of $\tilde{\Gamma}$ in Proposition 3: these directions undo the correlation between markets, so that their trend factors are, to first order, uncorrelated with unit variance and earn λ_k. The spectrum of a filter is then its vector of factor weights and its alignment the choice of factors: in the benchmark the spectrum fixes convexity, while at a given spectrum the alignment bounds the first-order Sharpe ratio (part (a), by von Neumann's trace inequality). The Sharpe-optimal filter C^+ is, to first order in the benchmark, the tangency portfolio of the trend factors with nonnegative weights $\vartheta_k \propto \lambda_k^+$: the optimal linear map of Brandt and Santa-Clara (2006), which Firoozye, Tan, and Zohren (2023) give in closed form and Kelly et al. (2023) as a Markowitz solution, restricted to positive semidefinite filters. Part (b) bounds the first-order Sharpe ratio at a given convexity breadth: more convexity than that of C^+ costs Sharpe ratio. The frontier is of leading order in each coordinate: the Sharpe ratio is of first order in ε, the skewness is evaluated in the benchmark, and alignment shifts the skewness at the same order in a direction-dependent way. New relative to this literature are the skewness of these filters and the separation of spectrum and alignment. Beyond the benchmark the premium also moves the third cumulant, along other directions, so the effect of alignment on skewness has no general sign (Remark B.2 in Section B of the Appendix). In the benchmark, moreover, skewness at a given spectrum does not depend on the directions: the leading eigen-trend carries the convexity of EQ because it carries the largest eigenvalue of EQ, and a filter that moves this weight to other directions keeps the skewness. The crisis payoff, by contrast, is tied to the direction of the risk-on/risk-off trend (Section 3.2), so the trade-off that matters for the investor is between the Sharpe ratio and the crisis payoff. Because the propositions order filters rather than predict the level of their Sharpe ratios, the second prediction, P2, is ordinal: at a given spectrum, alignment with the trend premium raises the Sharpe ratio, while whether alignment lowers skewness and whether the gain survives implementation are empirical questions.

Proposition 4 (Crisis payoff in the benchmark)**.** *Under Assumption 1, let C be a cross-sectional filter, a a portfolio of the normalized returns with $a^\top Pa > 0$, and $\check{z} = a^\top \sum_{t=1}^{T} z_t/(T\, a^\top Pa)^{1/2}$ its standardized T-day return. Then*

$$\mathbb{E}\big[F_T^C \mid \check{z}\big] = b_T\, \Lambda_C\, (\check{z}^2 - 1), \qquad b_T = \frac{1}{T}\sum_{h=1}^{T-1}(T-h)\beta_h, \qquad \Lambda_C = \frac{a^\top PCPa}{a^\top Pa}. \tag{11}$$

Λ_C is a Rayleigh quotient of $P^{1/2}CP^{1/2}$, whose eigenvalues are those of CP, in the direction $P^{1/2}a$; at a given spectrum it ranges between the smallest and the largest eigenvalue of CP, depending only on the directions of the filter.

The trend portfolio thus earns a quadratic payoff in any portfolio of the markets, the smile of trend following, and its payoff when that portfolio falls sharply is the exposure Λ_C of the filter's directions to it. A filter can therefore keep the skewness of EQ and lose its crisis payoff: in a three-market example, two filters with the same spectrum have the same skewness but crisis payoffs of 0.81 and 0.49 standard deviations, and an exact target on the conditional variance removes the skewness but keeps 92% of the crisis payoff (Section B.4 of the Appendix). Directions drawn at random at the spectrum of EQ keep on average 9% of its exposure to the equity portfolio of the 81 markets, so filters whose directions are mostly estimation noise lose crisis payoff mechanically.

4.2. Filter designs and estimation

The candidate filters form a ladder from the benchmark EQ to the estimated optimum. Each step changes one object of Section 4.1 and so tests one route by which cross-sectional design could improve the trade-off between the Sharpe ratio and convexity. All are estimated from a training sample, in which hats denote estimates: $\hat{M} = W^{1/2}\hat{P}W^{1/2} = \sum_k \hat{\mu}_k \hat{q}_k \hat{q}_k^\top$, $\hat{\Pi}$, and, for a direction u_k, the training mean $\hat{\pi}_k = u_k^\top \hat{\Gamma}_s u_k$, with $\hat{\Gamma}_s$ the symmetric part of the training cross-moment $\hat{\Gamma}$, and variance $\hat{v}_k$ of its trend factor.

The first step changes only the spectrum: the spectral tilt (TILT) keeps the principal

components as directions and reweights them,

$$C^{\text{TILT}} = W^{1/2}\Big(\sum_k \hat{\mu}_k^{\alpha-1}\hat{q}_k\hat{q}_k^\top\Big)W^{1/2}, \tag{12}$$

with $\alpha = 1$ for EQ, $\alpha > 1$ concentrating risk on the leading eigen-trends, $\alpha = 0.5$, with equal risk weights, the agnostic risk parity of Benichou et al. (2017), and $\alpha = 0$ the naive Markowitz filter $C = \hat{P}^{-1}$, which flattens the spectrum completely; the cases $\alpha = 0$ and 0.5 are building blocks of Valeyre (2023), who combines them linearly. These designs test whether convexity can be bought by concentrating risk on the common trends.

The second step keeps a fixed basis but weights it by the premium: the market-premium filter MKT and the principal-component filter (PCA) weight the markets or the principal components by premium over variance, $\omega_k \propto \hat{\pi}_k^+/\hat{v}_k$, with $\hat{\pi}_k^+ = \max(\hat{\pi}_k, 0)$, the tangency weights if the factors are uncorrelated, and so test whether tilting risk toward past premia raises the Sharpe ratio.

In the third step, the RP-PCA and SPCA filters choose the directions themselves from the premium. Both start from the eigen-trends, the eigenvectors of M, which select directions for variance alone, and they bring in the premium in different ways. The RP-PCA filter adapts the method of Lettau and Pelger (2020), which changes the matrix that is decomposed by adding to the second moments of returns extra weight on the outer product of mean returns. For trend P&L the premium of a direction q is the quadratic form $q^\top \Pi q$, so the premium matrix Π takes the place of that outer product. The algorithm estimates $\hat{M}$ and $\hat{\Pi}$ in the training window, takes the eigenvectors of $\hat{M}_\gamma = \hat{M}/\text{tr}(\hat{M}) + \gamma\,\hat{\Pi}/\text{tr}(\hat{\Pi})$ with $\gamma = 1$, which span the largest share of variance plus γ times the share of the trend premium and return the eigen-trends at $\gamma = 0$ (Proposition E.2 in Section E.1 of the Appendix), and trades them at premium-over-variance weights. The SPCA filter adapts the method of Giglio et al. (2025), which changes the sample that is decomposed; unlike theirs, whose screen is on the absolute covariance or correlation with an observed factor, the screen here is signed and on

the premium, and there is no target factor to project. In each of three steps the algorithm ranks the markets by the elements of $R_j\hat{\Pi}R_j\mathbf{1}$, their contributions to the premium of the trend factor on the residual equal-weight portfolio $R_j\mathbf{1}$ ($\mathbf{1}$ the vector of ones, R_j the projection off the directions already found), keeps the 20 largest, and takes the leading eigenvector of $R_j\hat{M}R_j$ restricted to them; the directions are traded at the same weights. RP-PCA thus keeps the whole panel and rotates the eigen-trends toward the premium, and tests whether directions chosen for premium as well as variance raise the Sharpe ratio; SPCA concentrates on a few premium-rich directions, and tests whether they do so with a concentrated, and therefore more skewed, spectrum.

The last step aligns the directions with the premium: the aligned filter and the plug-in optimal filter OPT estimate the whitened factors of Proposition 3 directly,

$$\hat{C}_\vartheta = \hat{P}^{-1/2}\Big(\sum_k \hat{\mu}_k \hat{\tilde{q}}_k \hat{\tilde{q}}_k^\top\Big)\hat{P}^{-1/2}, \qquad \hat{C}^+ = \hat{P}^{-1/2}\Big(\sum_k \hat{\lambda}_k^+ \hat{\tilde{q}}_k \hat{\tilde{q}}_k^\top\Big)\hat{P}^{-1/2}, \tag{13}$$

with $\hat{\mu}_k$ the spectrum of EQ on the estimation sample and $\hat{\lambda}_k$, $\hat{\tilde{q}}_k$ the eigenvalues and eigenvectors of $\hat{\tilde{\Gamma}} = \hat{P}^{-1/2}\hat{\Gamma}_s\hat{P}^{-1/2}$. $\hat{C}_\vartheta$ (labeled C_ϑ in the tables, as $\hat{C}^+$ is labeled OPT) isolates alignment at the spectrum of EQ, and $\hat{C}^+$, the estimated tangency portfolio, measures what an unrestricted premium estimate can deliver. The designs draw on eigen-portfolios and covariance filtering (Avellaneda and Lee, 2010; Pantaleo, Tumminello, Lillo, and Mantegna, 2011; Zumbach, 2011; Bongiorno and Challet, 2024); Section E.1 of the Appendix gives every estimator.

All filters share one protocol: the same markets and training window, a training correlation shrunk by 10% toward the identity, a fallback to EQ, and the variance of EQ as common scale. They are compared with EQ in three stages: (1) gross and out of sample, in two fixed splits (2001–2012 and 2013–2026, and the reverse) and in walk-forward designs re-estimated every year, on an expanding window from 2004 and on a 120-month window from 2011; (2) under the trading frictions of Section 2.3; and (3) net of costs, re-estimated monthly from

three years of data and traded as instrument positions, without and with turnover control. The same protocol on 1984–2000 tests the filters on data not used to develop them (Section E.4.2 of the Appendix). Table 6 gives the levels of the main designs and compares them with EQ and with each other.

Panel A: Levels

Filter	Gross Sharpe	Gross Skewness	Net, turnover control Sharpe	Net, turnover control Skewness	Crisis (%)	CE (%)	Risk in eigen-trend 1	Λ_C
Equal risk (EQ)	0.87	1.49	0.63	1.31	8.1	4.48	43%	0.132
Market premium (MKT)	1.24	1.41	0.68	1.31	6.0	4.51	29%	0.087
Risk-premium (RP-PCA)	1.17	1.29	0.83	0.95	7.4	4.77	22%	0.081
Supervised (SPCA)	0.69	2.28	0.57	1.60	4.1	4.22	40%	0.106
Plug-in optimal (OPT)	1.54	0.97	0.85	1.12	1.6	4.67	7%	0.013
Aligned C_ϑ	1.51	1.17	0.81	1.33	2.0	4.58	7%	0.015

Panel B: Differences to EQ, 90% intervals; bold: the interval excludes zero

Filter	Gross skewness	Net skewness	Crisis (pp)	CE (bp)
Market premium (MKT)	−0.08 [−0.30, 0.37]	0.00 [−0.50, 0.69]	**−2.1 [−3.3, −1.0]**	3 [−37, 46]
Risk-premium (RP-PCA)	−0.20 [−1.03, 0.57]	−0.36 [−1.37, 0.42]	−0.8 [−3.9, 2.4]	29 [−28, 85]
Supervised (SPCA)	0.79 [−0.35, 1.51]	0.29 [−0.45, 1.31]	**−4.0 [−6.4, −1.6]**	−26 [−76, 25]
Plug-in optimal (OPT)	−0.52 [−0.92, 0.23]	−0.19 [−1.39, 0.77]	**−6.5 [−10.6, −2.8]**	19 [−65, 101]
Aligned C_ϑ	−0.32 [−0.73, 0.39]	0.02 [−1.40, 1.06]	**−6.2 [−10.1, −2.6]**	10 [−68, 94]

Panel C: The filters against each other, 90% intervals; bold: the interval excludes zero

Pair	Gross Sharpe	Gross skewness	Net Sharpe	Net skewness	Crisis (pp)	CE (bp)
RP-PCA − OPT	**−0.37 [−0.69, −0.08]**	0.32 [−0.53, 1.07]	−0.03 [−0.33, 0.27]	−0.17 [−0.81, 0.59]	**5.7 [3.7, 7.7]**	10 [−67, 87]
RP-PCA − SPCA	**0.47 [0.13, 0.78]**	−0.99 [−2.26, 0.29]	0.25 [−0.04, 0.61]	**−0.65 [−1.34, −0.13]**	**3.2 [0.4, 6.4]**	54 [−16, 125]
OPT − SPCA	**0.85 [0.50, 1.22]**	**−1.31 [−2.09, −0.22]**	0.28 [−0.08, 0.67]	−0.48 [−1.30, 0.12]	−2.5 [−6.1, 1.1]	44 [−51, 140]

Exhibit 6: **Cross-Sectional Filters: Levels and Differences.** Gross: expanding walk-forward re-estimated every year, 2004–2026, 81 markets, scale-free P&L. Net: monthly re-estimation, net of costs, turnover control ($\eta = 0.1$), January 2001–October 2026, 10% volatility. Skewness: 55-day, widened intervals. Crisis: mean return in the 11 worst S&P 500 quarters. CE: certainty-equivalent return of a 20% sleeve on a 60/40 portfolio, relative risk aversion 5. Risk in eigen-trend 1: mean share of the position variance in the first eigenvector of the full-sample correlation matrix (real-time eigenvectors: Section C.3 of the Appendix). Λ_C: exposure of Proposition 4 to the equity portfolio, estimated before each crisis quarter from the preceding three years, mean over the quarters. Sharpe: stationary bootstrap, differences to EQ in Figure 7; crisis: bootstrap over the crisis quarters; CE: block bootstrap of months; intervals unadjusted. Designs: Section 4.2; all stages: Table A32 of the Appendix.

4.3. Spectrum-changing designs

The spectrum links a filter to convexity (Proposition B.2 in Section B of the Appendix), and the data confirm this link at the two ends, where the spectrum is concentrated or spread evenly. With positions fixed in one half of the sample, the filters that concentrate the spectrum, the SPCA filter in both splits and the spectral tilt in the forward split, are among the most skewed, with about two convexity bets or fewer. The naive Markowitz filter $C = \hat{P}^{-1}$ spreads the spectrum evenly and is the least skewed filter in every design, 0.56 to 0.66 against 1.16 to 1.49 for EQ. Between the two ends the ordering is weak, and

the premium-weighted filters, like EQ, are more skewed than the spectral law predicts in nine of ten cases (Section E.2 of the Appendix). In 1984–2000 the spectral tilt is again more skewed than EQ in both splits. The spectrum thus controls convexity at the extremes, where concentrating risk on the leading eigen-trends raises skewness and flattening the spectrum lowers it; raising the Sharpe ratio requires changing the directions.

4.4. Premium-aware directions and alignment

Using the premium, rather than the spectrum alone, targets the Sharpe ratio: premium weights and RP-PCA directions raise the gross Sharpe ratio when the premium is sufficiently persistent. In the expanding walk-forward, the market-premium filter gains 0.37 and the RP-PCA filter 0.29 over EQ, both with intervals above zero (Panel A of Figure 7). The SPCA filter, which concentrates on a few premium-rich directions, loses 0.18 and is the most skewed design. Out of sample the premium structure of RP-PCA weakens: estimated in 2001–2012, its first five directions carry 130% of the training premium, but in 2013–2026 only 74%, against 67% for the first five principal components (Section E.6.1 of the Appendix).

At a spectrum fixed on the estimation sample, aligned filters raise the gross Sharpe ratio. Re-estimated every year, C_ϑ raises it by 0.64 and 0.74 on the expanding and the rolling window, and OPT, the estimated optimum, gains most, 0.66 and 0.84; in the two fixed splits the gains of C_ϑ, 0.11 and 0.41, are not significant. On the correlations of the test sample the spectrum of C_ϑ is no longer that of EQ: its convexity breadth is 3.3 against 5.0 for EQ in one fixed split and 4.7 against 3.3 in the other, and in the median test year of the walk-forward 3.9 against 4.1 (Section E.3 of the Appendix). A test of superior predictive ability over all 33 reported walk-forward variants rejects the null that none beats EQ, also with the trading frictions and with every variant rescaled to the volatility of its benchmark, but the deflated Sharpe ratios of the gains of OPT and C_ϑ, 0.41 to 0.77, stay below the conventional 0.95 (Section F.2 of the Appendix). Skewness, by contrast, does not separate the filters: no filter changes the skewness of EQ significantly, a loss of 0.5 is excluded for MKT and SPCA but

not for RP-PCA, C_ϑ, and OPT, and the largest loss, -0.52 for OPT, has a sign that depends on how the yearly filters are sized (Section E.4.1 of the Appendix).

The gross gains do not isolate alignment with lead–lag premia, because weighting markets by their own premia (MKT) already gains 0.37. They also come at a price in crises. The aligned filters keep 7% of their position risk in the leading eigen-trend, against 43% for EQ (4% against 35% with real-time eigenvectors; Section C.3 of the Appendix), and their crisis payoff falls by 6.5 and 6.2 percentage points a quarter (90% intervals -10.6 to -2.8 and -10.1 to -2.6; Panel C of Figure 7); RP-PCA keeps most of it, and SPCA loses 4.0 (-6.4 to -1.6) with no gain to set against it. In the 14 bear episodes the plug-in optimal and the aligned filter gain 2.6% and 3.5% on average, against 11.1% for EQ, and OPT earns less than EQ in 13 of them (one-sided sign test $p = 0.001$; Table 3). The gross Sharpe gain of the aligned filters thus comes with leaving the trend that carries the crisis payoff, while the change in their skewness is not significant, as Propositions 3 and 4 imply for a change of directions at a given spectrum. In the data, Λ_C, with the equity portfolio as a and measured at the start of each of the 11 worst equity quarters from the preceding three years, correlates 0.85 with the net crisis payoff across the six filters (Spearman 0.77), and within quarters a higher exposure goes with a higher payoff (slope 0.31, 90% interval 0.02 to 0.69), against a correlation of 0.31 with net skewness (Section E.8 of the Appendix). Among EQ, MKT, RP-PCA, and SPCA the correlation is 0.24: a low exposure rules the payoff out, and a high exposure does not by itself secure it. Estimation noise reinforces the loss: the leading eigenvector of the estimated premium matrix has a median absolute cosine of only 0.40 with the leading eigen-trend in three-year windows (Section E.6.2 of the Appendix).

4.5. Evidence from 1984–2000

Because the filters were developed on 2001–2026, the earlier period tests whether the gross evidence carries over; it does not. On 1984 to 2000 and the 21 markets that trade then, the market-premium filter, OPT, and C_ϑ show no significant gain in either fixed split, and

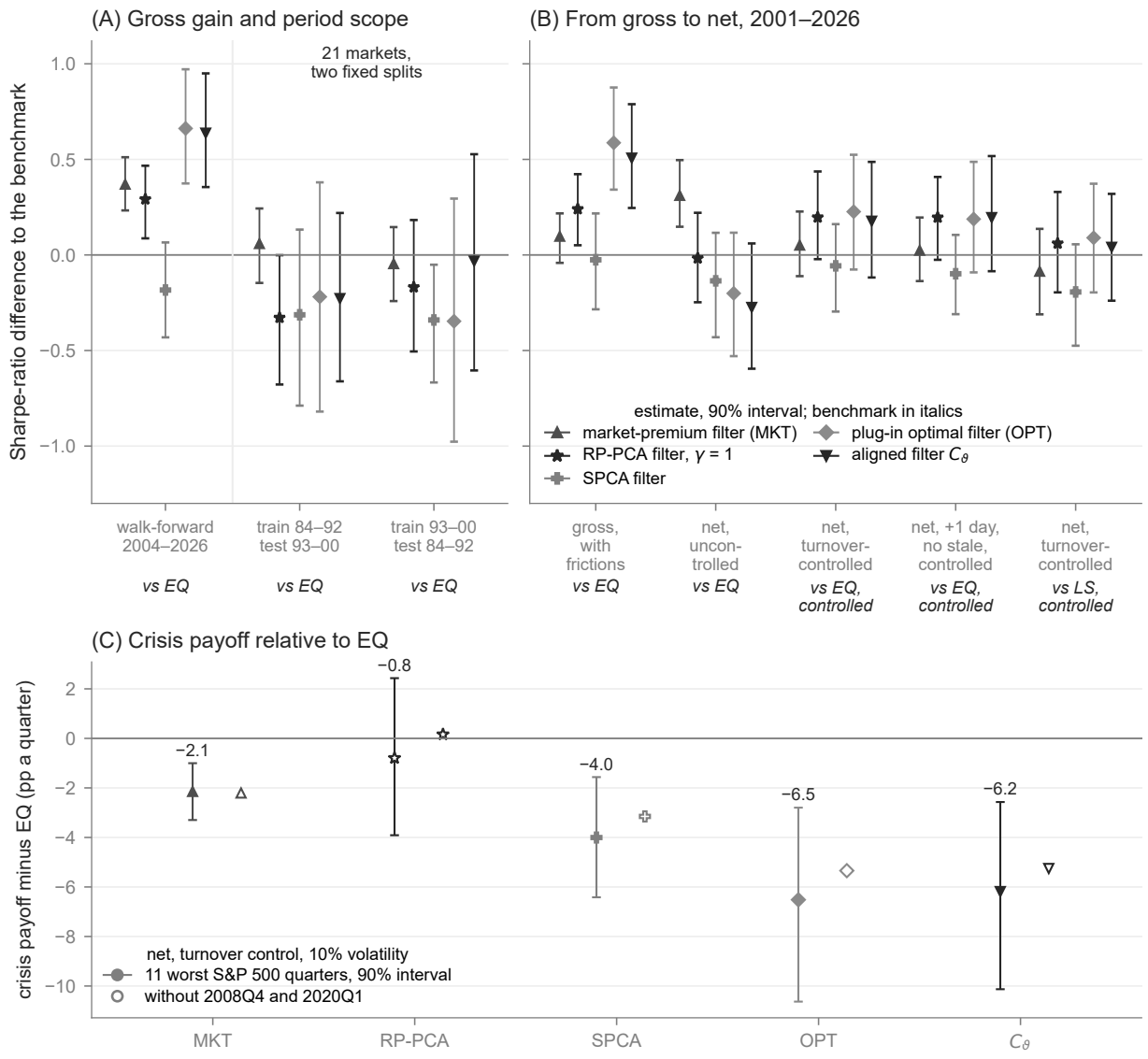


Exhibit 7: **Cross-Sectional Filters from Gross to Net Performance.** A point above zero beats the benchmark named below the stage; with turnover control, every interval in Panel B includes zero. 81 futures markets, January 2001–October 2026, and 21 markets, 1984–2000; Sharpe-ratio differences to the benchmark named below each stage, 90% stationary-bootstrap intervals. Panel A: gross, annual expanding walk-forward 2004–2026, and the two fixed splits of 1984–2000. Panel B: gross with the trading frictions (one more day, cross-market terms lagged, no stale markets); net, monthly re-estimation, without and with turnover control ($\eta = 0.1$); controlled, one more day, no stale markets; controlled against LS(250,20). Panel C: crisis payoff relative to EQ, net, controlled, 10% volatility, bootstrap over the 11 crisis quarters; with intervals: all quarters, filled if the interval excludes zero; offset hollow: without 2008Q4 and 2020Q1.

the RP-PCA and the SPCA filter lose significantly in one fixed split each, by 0.33 and 0.34 (Panel A of Figure 7). For OPT the upper bounds of both early intervals, 0.38 and 0.29, lie below its later walk-forward gain of 0.66, so the early data do not support a gain of the

later magnitude (Section E.4.2 of the Appendix). The gross evidence for premium-aware and aligned filters is specific to the later sample.

4.6. From gross to net performance

A gross gain improves the investor's trade-off only if it survives implementation, and estimation, turnover, and costs each absorb part of it (Panel B of Figure 7). Trading frictions separate the filters first: OPT keeps at least 0.46 in each of the 14 friction cells, with intervals above zero and Romano–Wolf adjusted p-values of at most 0.05 in every cell of the adjusted family, whereas the market-premium filter keeps only 0.10, consistent with short-horizon autocorrelation of single markets (Section E.4.3 of the Appendix). Monthly re-estimation from a three-year window then lowers the gross gains of OPT and C_ϑ from 0.66 and 0.64 to 0.41 and 0.37 before any cost is charged (Section E.5 of the Appendix).

Net of costs, OPT and C_ϑ turn over about 490 and 510 times capital a year, against 204 for EQ, and fall behind it, whereas MKT gains 0.31 (0.15 to 0.50) over the uncontrolled EQ, a gain that disappears once EQ also trades more slowly. Every portfolio is therefore also traded with turnover control: positions move each day only part of the way toward the target positions H_t^*, $H_t = H_{t-1} + \eta(H_t^* - H_{t-1})$, here at $\eta = 0.1$ (10% a day) (Gârleanu and Pedersen, 2013). Turnover falls by about three quarters and the controlled EQ earns 0.63 instead of 0.39. Against it, OPT, C_ϑ, and RP-PCA gain 0.23, 0.18, and 0.20, MKT gains 0.05, and SPCA loses 0.06, none significantly (Panel B of Figure 7); against the controlled LS(250,20) the gains are 0.09 or less. Turnover control thus removes the cost penalty without producing a statistically significant net gain.

The same holds under the ex-ante rule, at other adjustment speeds, with doubled costs, and under the base-case frictions, where OPT and C_ϑ earn 0.19 and 0.17 less than the controlled LS(250,20); at the standard and at halved costs the aligned filters lead the reference strategy in point estimate. The investable base case, the controlled EQ under the base-case frictions, earns a net Sharpe ratio of 0.36 without a portfolio rule and 0.49 with the ex-ante

rule, against 0.54 and 0.76 for the controlled LS(250,20) (Sections E.5.2 and E.8.3 of the Appendix).

4.7. Summary of the cross-sectional evidence

The answer to the second question has two parts. First, the evidence matches the theory: the spectrum moves convexity at its extremes, and MKT, RP-PCA, and the aligned filters raise the gross Sharpe ratio out of sample from 2001 on, while no filter changes skewness significantly. Second, the premium structure that the theory exploits is too weak, relative to its estimation error (Kan and Zhou, 2007; Kozak, Nagel, and Santosh, 2020) and trading cost, to be captured reliably: in three-year windows only the leading eigenvalue of the estimated premium matrix exceeds a 95% noise band, in 15 of 23 windows (Section E.6.2 of the Appendix); 1984–2000 does not confirm the gross gains; and monthly re-estimation, turnover, and costs leave no statistically significant net gain over the controlled EQ or the reference strategy. For the research question, the designs span the trade-off. SPCA holds the most skewness, 2.28 gross against 0.97 for OPT, at a Sharpe ratio 0.85 lower. OPT and C_ϑ trade crisis payoff for gross Sharpe ratio, because their estimated directions are not the ones that carry the crisis payoff. RP-PCA lies between them and is the most balanced net of costs: against OPT its crisis payoff is 5.7 percentage points a quarter higher (3.7 to 7.7) at a Sharpe difference of -0.03 (-0.33 to 0.27) (Section E.8 of the Appendix).

5. Risk targeting and delivered convexity

Cross-sectional design determines where the portfolio takes its convexity; the portfolio-level risk rule determines how much of it appears in measured returns. The third part of the research question is therefore how much of the available convexity a risk rule preserves, and at what Sharpe ratio. An exact target on the conditional variance removes the martingale part of convexity, the part generated by unexpected P&L, whereas estimated rules remove less, depending on their design and trading speed.

5.1. Conditional-variance targeting in theory

Both channels of Section 3.3 run through the conditional variance of the positions, the object that a portfolio risk rule controls.

Corollary 2 (Risk targeting)**.** *Under Assumption 2, let the positions be scaled by an $\mathcal{F}_{t-1}$-measurable factor $\ell_t > 0$ with $\sup_t \mathbb{E}|\ell_t g_t|^3 < \infty$ and $\sup_t \mathbb{E}|\ell_t e_t|^3 < \infty$, so that the portfolio earns $\ell_t g_t$, with innovation $\ell_t e_t$ and martingale sum $E_T^\ell = \sum_t \ell_t e_t$. (a) $\kappa_3(E_T^\ell) = 3\sum_{s<t} \mathrm{Cov}(\ell_t^2 V_t, \ell_s e_s)$. (b) If $V_t > 0$ almost surely and the conditional variance is targeted, $\ell_t^2 V_t = \bar{V} > 0$, then $\kappa_3(E_T^\ell) = 0$: the scaled position and co-movement channels cancel, and the third cumulant of the horizon return reduces to the term Δ_T of Proposition 2(b) computed for the scaled portfolio. (c) If instead $V_t^{\mathrm{pos}} > 0$ almost surely and the variance at a static correlation is targeted, $\ell_t^2 V_t^{\mathrm{pos}} = \bar{V}$, then $\kappa_3(E_T^\ell) = 3\sum_{s<t} \mathrm{Cov}\big(\ell_t^2 (V_t^{\mathrm{var}} + V_t^{\mathrm{cor}}), \ell_s e_s\big)$: the position channel vanishes and the co-movement channel of the scaled portfolio remains, with the deviations $V_t^{\mathrm{var}} + V_t^{\mathrm{cor}}$ weighted by $\ell_t^2 = \bar{V}/V_t^{\mathrm{pos}}$.*

The martingale part of convexity is the tendency of risk to rise after gains, so a rule that holds risk constant removes this tendency for every filter, and rules differ in how closely they approach this target. A rule that forecasts the risk of today's positions from an estimated covariance, the ex-ante rule, approximates part (b) (related correlation-aware rules: Baltas, 2015; Baltas and Kosowski, 2020); a rule on the realized volatility of past P&L, the ex-post rule, reacts with a lag and leaves part of the channels in place; and a static-correlation target approximates part (c), with a matrix $\hat{P}_t$ re-estimated each year. Hoffman and Kaminski (2016) argue that time-varying risk is a main source of trend skewness, that trend following at constant risk loses it, and that skewness and crisis protection are distinct outcomes; part (b) formalizes this argument for the martingale part. New here are the static-correlation target and the matched test below. The portfolio-level step of Harvey et al. (2018), which they read as an adjustment for time-varying correlations between assets, is an ex-post rule in these terms. Volatility targeting raises the Sharpe ratios of factor portfolios (Moreira

and Muir, 2017; Grobys, Ruotsalainen, and Äijö, 2018), risk assets (Harvey et al., 2018), and time-series momentum (Kim, Tse, and Wald, 2016), where risk adjustment can make skewness more negative (Dudler, Gmür, and Malamud, 2015), although its out-of-sample gains are contested (Cederburg, O'Doherty, Wang, and Yan, 2020; Bongaerts, Kang, and van Dijk, 2020); Masturzo (2025) finds that a short asset-volatility window all but removes the skewness of slow signals.

The third prediction, P3, is that a rule that holds the conditional variance of the positions constant removes the martingale part of the third cumulant for every filter; the drift part remains and has no general sign. If it is small relative to the removed part, as in the static benchmark where it is zero, the daily ex-ante rule, which approximates that target, should remove most of the measured skewness; whether the drift part of the scaled portfolio is small is an empirical premise, and the rule targets $\mathbb{E}[g_t^2 \mid \mathcal{F}_{t-1}] = V_t + d_t^2$, which differs from V_t at second order in the drift.

Six portfolio risk rules are compared, each applied to the positions of the portfolios of Table A49. Without a portfolio rule, the positions are scaled to 10% volatility by one full-sample constant. The *ex-ante* rule scales the positions decided on day t by the inverse of their volatility forecast from an EWMA covariance (span 60) estimated through $t-1$, and so approximates the target of Corollary 2(b). The two *ex-post* rules scale them by the inverse of the lagged realized volatility of the P&L of the target positions (span 100, the primary comparator, or 60). The *static-correlation* target uses their ex-ante volatility at the correlation matrix of all previous years, the target of Corollary 2(c), and *partial targeting* scales by the ex-ante volatility forecast to the power -0.5, between no rule and the ex-ante rule. Each rule is traded at its target daily ($\eta = 1$) or with partial adjustment at $\eta = 0.25$, 0.1, or 0.05 (Section G.2.10 of the Appendix). All rules act on positions, not on the P&L after the fact, use information through $t-1$, and pay the costs of the resulting trades. The *matched comparison* holds the speed fixed and replaces the ex-post rule (span 100) by the ex-ante rule, so that both rules trade with the same partial adjustment.

5.2. Daily targeting and measured skewness

Traded every day, the ex-ante rule removes most of the measured 55-day skewness of the mechanism benchmarks, the portfolios without turnover control (Exhibit A20). Without a rule, the 55-day skewness is 1.38 for the implementable EQ and 0.55 for LS(250,20); the daily ex-ante rule leaves 0.07 and 0.13, differences of −1.31 (widened interval −1.94 to −0.66) and −0.42 (−0.70 to −0.16). The skewness of compounded calendar-quarter returns falls from 1.56 to 0.31 and from 0.71 to 0.31 (Section D.2.4 of the Appendix). For the scale-free EQ, whose skewness the daily ex-ante rule lowers from 1.25 to −0.01, the decomposition leaves a martingale part of 0.12 and a drift part of −0.14 (Panel C of Figure 5), close to the zero martingale part that Corollary 2(b) implies for an exact target. A placebo without a trend signal shows that part of the fall is induced by the rule itself: the rule lowers the 55-day skewness of positions with random monthly signs from 0.00 to −0.23. The result holds with capped leverage, without the stale markets or the short-term interest-rate futures, for the spectral tilts of EQ, and in 99 of 100 random universes of 40 markets; on unadjusted contracts of the 42 commodities the ex-ante rule keeps 14% of the skewness without a rule and the ex-post rule 47% (Sections D.2 and D.3 of the Appendix).

Skewness and crisis payoff are distinct outcomes. Under the daily ex-ante rule the mean return in the worst decile of S&P 500 quarters falls from 8.9% to 7.2% for EQ (−1.8 percentage points, −5.3 to 0.4) and from 6.9% to 6.6% for LS(250,20) (−0.4, −2.4 to 1.1), and the smile coefficient of EQ falls from 2.13 to 1.28 (−0.84, −1.51 to 0.27); none of these changes is significant, although in the fourth quarter of 2008 alone EQ earns 10.2% under the rule against 26.5% without it (Sections D.2.3 and D.2.7 of the Appendix). Rescaled to equal realized volatility, the maximum drawdown of EQ deepens from −35% to −52% (Section D.2.6 of the Appendix), a loss from 2015 that it has not recovered by the end of the sample, mostly in equities and government bonds, while that of LS(250,20) hardly changes. A fast volatility target thus removes most of the time-series skewness for which trend-following al-

locations are typically held, while the crisis payoff and the smile do not change significantly. With partial adjustment ($\eta = 0.1$) the ex-ante rule lowers the skewness of LS(250,20) only from 0.43 to 0.30 (-0.37 to 0.14), which is why the comparison below holds the speed fixed.

5.3. Trading speed and a matched comparison

The trade-off depends on the design and the trading speed of the rule (Figure 8). Traded daily, the ex-ante rule lowers the net Sharpe ratio of the implementable EQ from 0.39 to 0.28 (difference -0.24 to 0.03), because turnover rises from 204 to 298 times capital a year, and raises that of LS(250,20) from 0.74 to 0.91 (0.06 to 0.27). The ex-post rule (span 100), which scales by the realized volatility of past P&L, keeps more skewness than the daily ex-ante rule, 0.26 more for LS(250,20) (0.14 to 0.45) and 0.66 for EQ (0.31 to 1.08; Section D.2.5 of the Appendix), at a net Sharpe ratio that differs by 0.05 or less for LS(250,20) and is 0.15 higher for EQ (0.01 to 0.29); the static-correlation target keeps slightly more. The daily ex-ante rule holds the realized 63-day volatility of both portfolios between 8% and 12% on 82% of days, against 53% and 61% for the ex-post rule and about a third without a rule (Section C.6 of the Appendix). Slower adjustment raises the net Sharpe ratio of every rule, so a comparison of designs must hold the trading speed fixed.

The matched comparison isolates the design of the rule from its speed. At $\eta = 0.1$ the ex-ante rule raises the net Sharpe ratio of LS(250,20) by 0.10 (0.04 to 0.17) over the ex-post rule, 0.99 against 0.89, and by 0.23 (0.12 to 0.33) over no rule (Exhibit A21; Panels C and D of Figure 8). The gain over the ex-post rule keeps its sign with one-year bootstrap blocks and under a HAC test ($p = 0.03$), survives a Romano–Wolf adjustment over the 18 cells of rule, speed, and comparator ($p = 0.04$), and holds under the base-case frictions, 0.11 (0.04 to 0.18), in both halves of the sample, with covariance spans of 30 and 100 days, and when only the risk scale moves 10% a day toward its target while the signal positions trade daily, 0.10 (0.03 to 0.17) at a crisis payoff of 7.1% against 6.6% (Section D.2.5 of the Appendix). Its skewness differs by only -0.11 (-0.26 to 0.03) from the 0.41 of the slow ex-post rule. A

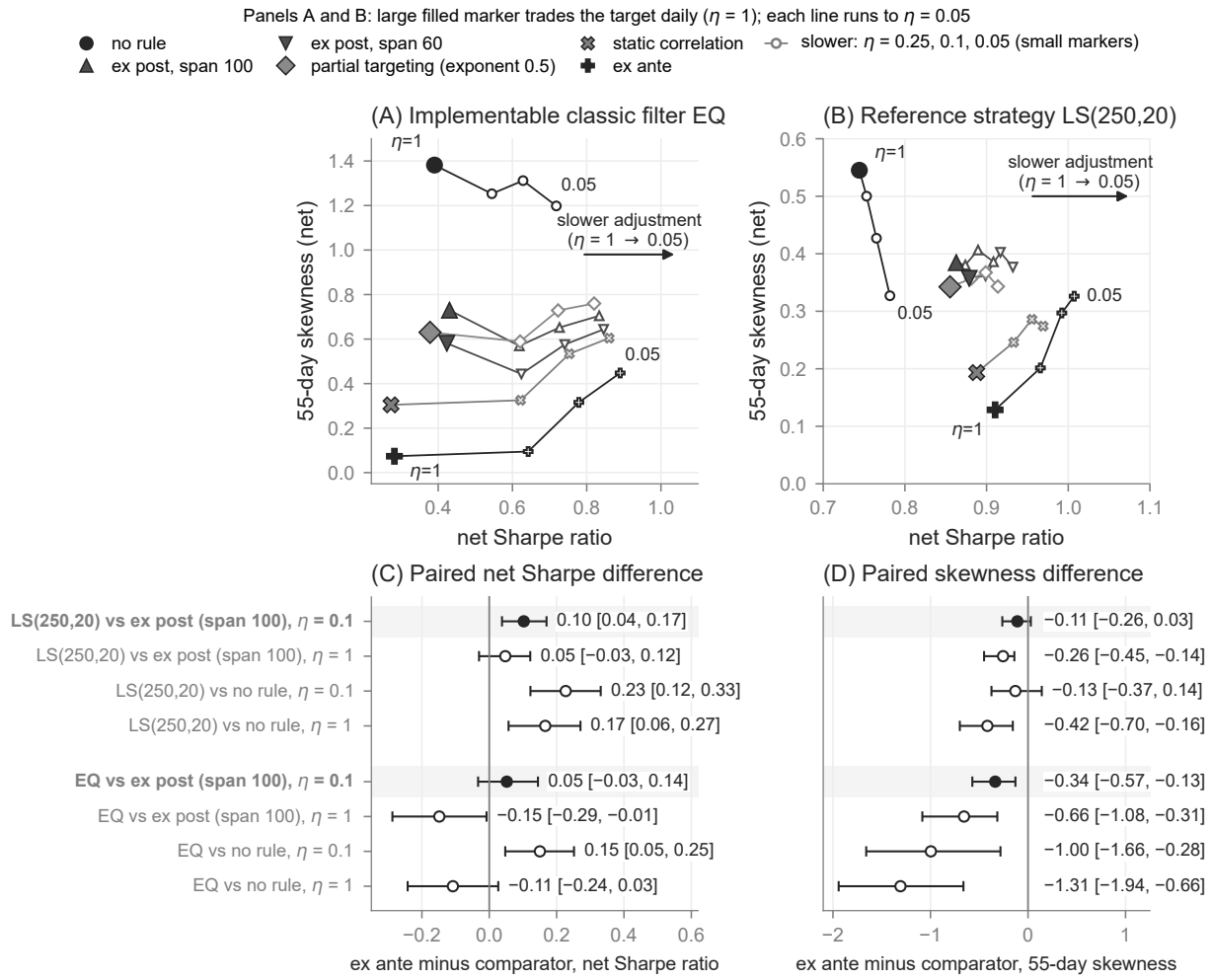


Exhibit 8: **Portfolio Risk Rules: Speed, Sharpe Ratio, and Skewness.** Traded daily, the ex-ante rule has the lowest skewness; slower trading moves every rule to a higher net Sharpe ratio. 81 futures markets, January 2001–October 2026; net of costs, 10% volatility. Panels A (implementable EQ) and B (reference strategy LS(250,20)): net 55-day skewness against net Sharpe ratio under the six rules of Section 5.1; large markers trade the target daily ($\eta = 1$), lines run through $\eta = 0.25$, 0.1, 0.05 (small markers). Panels C and D: ex-ante rule minus ex-post rule (span 100) or no rule at equal speed, with 90% intervals (Sharpe: stationary bootstrap; skewness: widened), so positive values favor the ex-ante rule; filled and shaded: the matched comparison at $\eta = 0.1$. Values: Exhibits A19 and A21.

slow rule moves the positions only partly toward the constant-risk target and so leaves more of the channels in place, while a forecast from current positions reacts without the lag of a rule on past P&L. For EQ the ex-ante rule gains no significant Sharpe ratio over the ex-post rule, 0.05 (−0.03 to 0.14), and lowers skewness at every speed.

5.4. Summary of the risk-targeting evidence

The third prediction, P3, holds, and it answers the third question: traded daily, the ex-ante rule leaves 0.07 of the 1.38 of the implementable EQ and 0.01 of 1.09 under the base-case frictions (Section E.5.2 of the Appendix). For the reference strategy, slow ex-ante targeting raises the net Sharpe ratio over a matched slow realized-volatility target, with little change in skewness; for the equal-risk portfolio the ex-ante rule lowers skewness at every speed. In the 14 bear episodes the slow ex-ante rule keeps 9.3% against 11.1% for EQ and 6.9% against 8.7% for LS(250,20) without a rule, as much as the matched ex-post rule (8.9% and 6.6%); it loses mainly in the fast sell-offs of November 2008 and March 2020, in which a rule that cuts positions as volatility jumps scales down the trend before it pays (Table 3). For the research question, the risk rule decides how much of the available convexity is measured and kept, and its design and speed are portfolio-construction choices, not implementation details.

6. Discussion

Sections 3 to 5 answer the research question in three steps: the spectrum fixes how much convexity diversification offers, cross-sectional design determines where the portfolio takes it, and the portfolio risk rule shapes how much of it appears in measured returns. To complete the answer, this section measures their economic magnitude, draws the implications for portfolio construction, and reports the inference and the scope of the evidence.

6.1. Economic magnitude

Figure 9 shows the three answers on the same axes of net Sharpe ratio and crisis payoff, one panel each. Their economic weight is measured for an investor who holds trend following at 10% volatility, alone or as a 20% sleeve, rebalanced monthly, of a 60/40 portfolio of S&P 500 and ten-year Treasury futures rebalanced quarterly (the benchmark of Sepp and Lucic, 2026, Section 7), in the tradition of crisis-alpha studies (Greyserman and Kaminski,

2014; Hamill, Rattray, and Van Hemert, 2016; Harvey, Hoyle, Rattray, Sargaison, Taylor, and Van Hemert, 2019). The measures are the mean return in the 11 worst of 103 equity quarters, the maximum drawdown, and the certainty-equivalent return (CE) at a relative risk aversion of 5, split into its mean-variance value and the remainder beyond this second-order approximation (Section E.8 of the Appendix). All returns are futures excess returns, without collateral interest. The 60/40 portfolio alone has a CE of 3.42% a year and a maximum drawdown of −30.6%.

Breadth beyond a dozen markets adds little skewness but retains economic value: net of costs, the median portfolio of 12 markets gives the sleeve a CE of 4.17%, all 81 markets 4.48%. The remainder beyond the mean-variance value is −3 to −1 basis points in every design, so the value of trend following runs through its Sharpe ratio and its hedge in equity sell-offs, not through the skewness of the sleeve.

Cross-sectional alignment trades crisis payoff for mean return. Net of costs and with turnover control, the plug-in optimal and the aligned filter earn 8.5% and 8.1% a year against 6.3% for EQ, but 1.6% and 2.0% in the worst equity quarters against 8.1%, with deeper drawdowns (−32% against −25%). The RP-PCA filter keeps the crisis payoff (−0.8, −3.9 to 2.4), and the SPCA filter lies below EQ on both axes (−4.0, −6.4 to −1.6), with a CE of 4.22% against 4.48%. In the 60/40 portfolio the two effects nearly offset: the CE gains over EQ, 19 and 10 basis points for OPT and C_ϑ and 29 for RP-PCA, have intervals that include zero.

The portfolio risk rule has the most precisely estimated economic effect. At $\eta = 0.1$, slow ex-ante targeting raises the CE of a LS(250,20) sleeve by 20 basis points a year over the matched ex-post rule (90% interval 6 to 34), with a crisis payoff of 5.7% against 5.2%; for EQ the gain is 11 basis points (−10 to 30). The gain comes from the higher mean, not from convexity. After a 2% management and a 20% incentive fee, every sleeve still raises the CE of the 60/40 portfolio in the main specification (3.93% to 4.42% against 3.42%), and

slow ex-ante targeting keeps 16 basis points. Under the base-case frictions the levels fall and the aligned filters lose their lead over EQ: they lose their crisis payoff entirely (−0.3% and 0.3% a quarter against 6.8% for EQ), slow ex-ante targeting keeps its edge for LS(250,20) (0.76 against 0.65), and after fees only the reference strategy still raises the CE of the 60/40 portfolio clearly (4.03%), whereas EQ leaves it unchanged and OPT and C_ϑ lower it (Sections E.8.2 and E.8.3 of the Appendix).

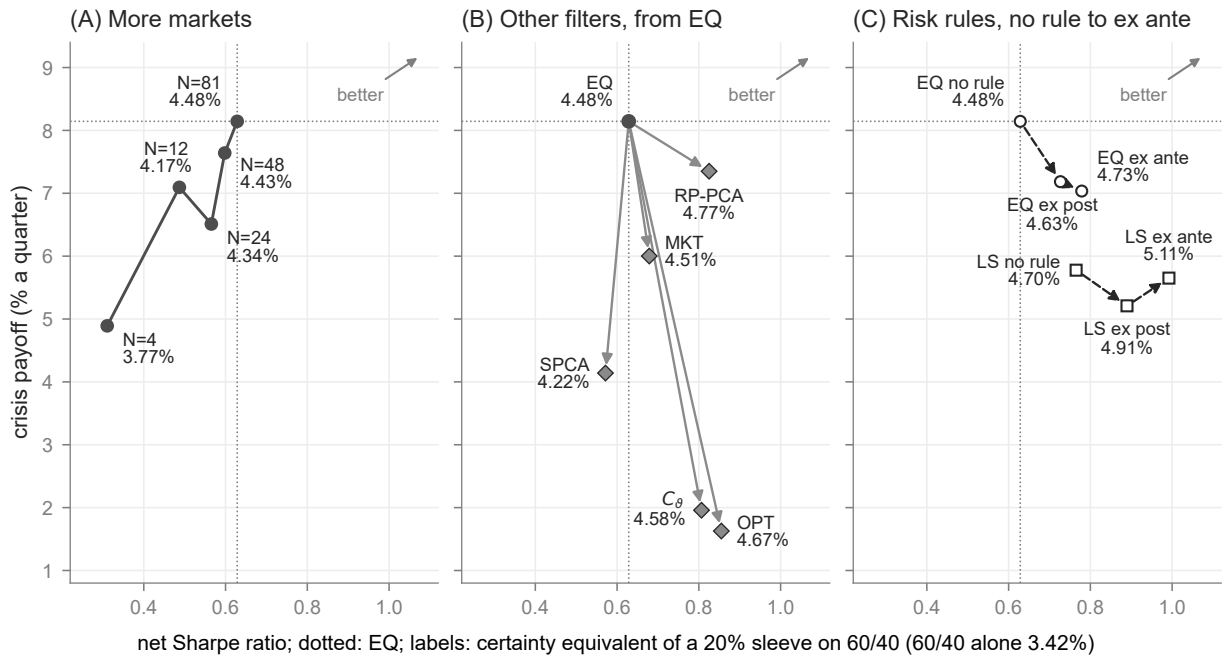


Exhibit 9: **Sharpe Ratio, Crisis Payoff, and Investor Value of the Design Choices.** Each panel changes one design choice; up and to the right is better, and the dotted lines mark the equal-risk portfolio EQ. More markets broadly raise both measures; the filters lose crisis payoff, most of them for a higher Sharpe ratio; the ex-ante rule raises the Sharpe ratio over the ex-post rule at a similar crisis payoff. 81 futures markets, January 2001–October 2026; net of costs, turnover control ($\eta = 0.1$), 10% volatility. Crisis payoff: mean return in the 11 worst S&P 500 quarters. Labels: certainty equivalent (% a year, relative risk aversion 5) of a 60/40 portfolio with a 20% trend sleeve. Panel A: EQ on random subsets of N markets (median of 30). Panel C: at $\eta = 0.1$, from no rule to the ex-post rule (span 100) to the ex-ante rule, for EQ and LS(250,20) (LS). Values: Section E.8 of the Appendix.

6.2. Implications for portfolio design

Four implications for portfolio construction follow, ordered along the three parts of the research question and then by implementation. First, diversification should be measured by moment: the number of markets and variance-based effective numbers of bets describe the

Sharpe ratio, while convexity is governed by the convexity breadth, which needs only the correlation matrix; twelve random markets keep the convexity of 81 but lose about 0.14 of net Sharpe ratio and 1 percentage point of crisis payoff. Second, a design that leaves the leading eigen-trend loses the crisis payoff, although keeping it does not by itself secure one, so the exposure should be checked: with positions, the share of position risk in that direction, $\hat{\mu}_1(\hat{q}_1^\top x_t)^2 / \sum_k \hat{\mu}_k(\hat{q}_k^\top x_t)^2$ for positions in normalized units, $x_{i,t} = \hat{\sigma}_{i,t} H_{i,t}/\sigma^*$, and eigenvectors estimated from past data (35% for EQ, 4% for OPT and C_ϑ); without positions, the correlation of a program's returns with an equity-minus-bond trend factor, which correlates with crisis payoff across the eleven implementable portfolios (Pearson correlation 0.91, Spearman 0.75, and 0.68 and 0.55 without OPT and C_ϑ; Section E.8.3 of the Appendix). Third, the risk rule is part of portfolio construction: a daily target on current positions removes most of the measured skewness at an average leverage 1.41 times that without a rule (Section D.2.6 of the Appendix), whereas slow ex-ante targeting raises the net Sharpe ratio of the reference strategy at the crisis payoff of a matched realized-volatility rule. Fourth, turnover control and frictions move the net Sharpe ratio of the equal-risk portfolio as much as any design (0.39 to 0.63 with control, 0.36 under the base-case frictions).

The results do not show that skewness itself has value for a CRRA investor (Section 6.1). The attributions to eigen-trends and asset classes are accounting identities, not causal effects, and the leading role of the risk-on/risk-off trend belongs to a sample in which stocks and bonds were negatively correlated before every crisis quarter (Section E.8.3 of the Appendix).

6.3. Statistical inference, robustness, and scope

The analysis reports medians and interquartile ranges across many portfolios and uses several resampling devices (Sections D.1 and F of the Appendix). A *moving-block bootstrap* (Künsch, 1989) with one-year blocks gives the skewness intervals, widened to the 1st to 99th percentiles (calibration in Section F.5 of the Appendix); conclusions rest on differences. A *stationary bootstrap* (Politis and Romano, 1994) with a mean block of 63 days gives the Sharpe intervals,

with a HAC test (Ledoit and Wolf, 2008) as check. Tests of superior predictive ability (Hansen, 2005), deflated Sharpe ratios (Bailey and López de Prado, 2014), and Romano–Wolf adjusted p-values (Romano and Wolf, 2005) address multiplicity, and *phase-randomized surrogates* (Prichard and Theiler, 1994) keep every auto- and cross-covariance of returns but destroy nonlinear dependence. The deflated Sharpe ratios of the fixed splits are also computed with each of the 235 reported Sharpe ratios of filter variants as a trial (Section F.2 of the Appendix).

The walk-forward gains of OPT and C_ϑ remain distinguishable from noise that keeps each market's first-order autocorrelation ($p = 0.01$ and 0.02), but no filter passes both this null and the deflation of its gain. The breadth ordering and the effect of the daily ex-ante rule hold when any asset class is left out, under a quantile measure of skewness, with winsorized P&L, and at other horizons (Section F.3 of the Appendix).

The scope of the results follows from the design. The horizon-free factorization holds for one signal kernel in every market, so different signal speeds, lead–lag terms, or capped signals lie outside its exact form; the channel split is an accounting identity; and Proposition 2 relaxes the Gaussian benchmark but does not cover skewed innovations such as the 2015 franc jump. All results rest on one panel of futures (Section G.1 of the Appendix).

7. Conclusion

This article asks how many independent bets a diversified trend follower holds on convexity, where these bets sit, and whether portfolio design can improve the trade-off between the Sharpe ratio and convexity.

The answer starts with a count. The 81 futures markets offer about 18 effective bets on the Sharpe ratio but only about 4 on convexity, and fewer when equity volatility is high. The reason is concentration: one risk-on/risk-off trend carries three quarters of the convexity in the spectrum, and with the next two eigen-trends it earns most of the crisis payoff.

Portfolio design acts on these few trends. In the benchmark, the spectrum sets the convexity of a portfolio, and at a given spectrum its directions set the crisis payoff. Filters aligned with estimated trend premia raise the gross Sharpe ratio from 2001 on but give up most of the crisis payoff. SPCA holds the most skewness, the optimum the highest Sharpe ratio, and RP-PCA keeps the crisis payoff at nearly the net Sharpe ratio of the optimum. The gross gains are not confirmed in 1984–2000, and after estimation, turnover, and costs, the filters differ in which of these properties they keep, not in a significant Sharpe gain.

The risk rule acts on what is measured. A daily target on the risk of current positions removes most of the measured skewness. A slow ex-ante target raises the net Sharpe ratio of the reference strategy over a matched realized-volatility rule, also after an adjustment for multiple testing, and keeps the crisis payoff of that rule. Other signals (Zakamulin and Giner, 2020), other universes, and returns after the end of the sample are natural tests of these findings.

A trend follower can diversify its Sharpe ratio across many markets, but its convexity is a bet on a few common trends, and its crisis payoff on the same trends, led by the risk-on/risk-off trend in bonds and equities. Investors who hold trend following for crisis protection should first check that a design has not left this trend, which a simple equity-minus-bond trend factor reveals, and then judge the design and speed of its risk rule.

Conflict of interest

The author is employed by Quintet Private Bank (Europe) S.A.; the views expressed are the author's own and do not necessarily reflect those of Quintet Private Bank.

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